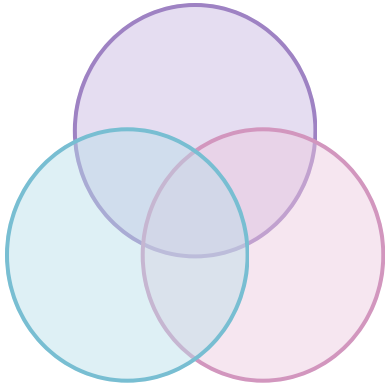


https://agencyai.mit.edu

agency *with* AI

Effective AI use in applied machine-learning projects

Lecture 1: What This Course Is

Shen Shen

Sep 15, 2026

2:30pm, Room 37-212

Slides in PDF

What motivated this course

- Over the years, recurring questions we get about 6.390:
  - why so nitty-gritty, so much math, so "bashy" assignments
  - why not more applied ML (a few of our own staff agree)

- For a long time, our reasoning:
  - 6.390 is about the fundamentals
  - without the nitty-gritty math, we can't understand the system
  - without coding and knowing the bashy gotchas, we can't spin up a proper test
  - (not to mention math and coding are fun!)

- Now, I think AI fluency and agency with AI are increasingly fundamental too:
  - from my own AI use, and what I see in students, staff, and colleagues
  - from how the field has been moving

When I first started using agentic AI, it was for routine tasks

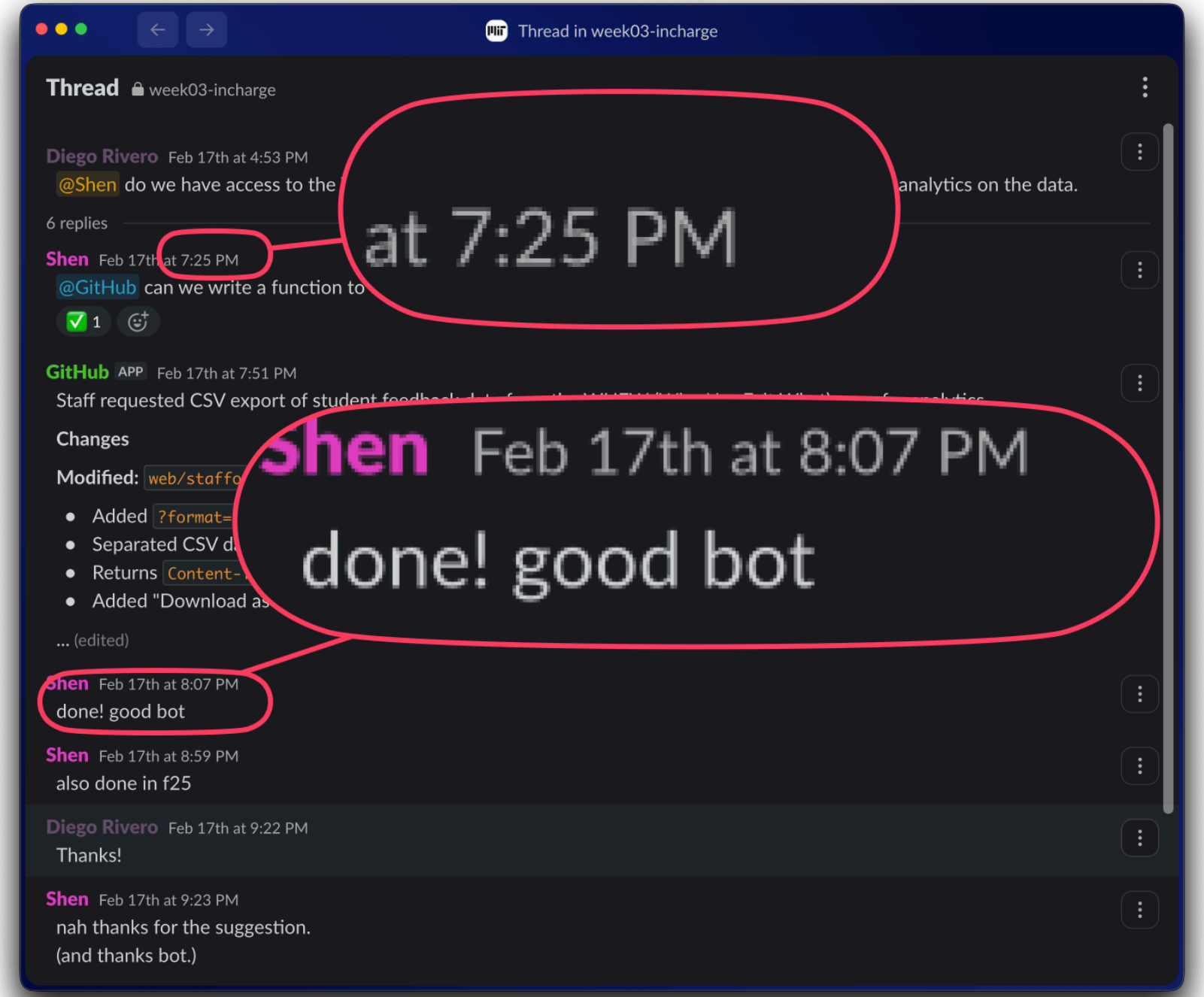
	kerb	feedback_length	feedback_difficulty	feedback_comment
0				
1		Just right	Just right	
2		Just right	Just right	The organization of question 4.2 was nice: specifically, it was helpful to have the "Potential Explanations" section at the end of the questions. Our group didn't look at the explanation choices as we were formulating our answers to 4.2.1-4.2.3, but it was great to be able to compare how we expressed our understanding in our own words to the listed explanations after we'd attempted each question.
3		Too Long	Too Hard	
4		Just right	Just right	
5		Just right	Just right	
6		Just right	Just right	
7		Just right	Just right	: -)
8		Just right	Just right	
9		Just right	Just right	
10		Just right	Just right	
11		Just right	Just right	
12		Just right	Just right	
13		Just right	Just right	
14		Too Short	Too Easy	
15		Just right	Just right	
16		Just right	Too Hard	
17		Just right	Too Easy	I don't like talking to people.
18		Just right	Just right	Thank you!
19		Just right	Just right	
20		Just right	Just right	
21		Just right	Just right	
22		Just right	Just right	
23				
24				
25				
26				
27				
28		Just right	Too Easy	
29		Just right	Too Hard	
30		None	None	
31		Just right	Just right	Good intro lab
32		Too Long	Just right	
33		Just right	Just right	
34		Just right	Just right	Great intro lab.
35		Too Long	Too Hard	
36		Just right	Too Easy	
37		Just right	Just right	
38		Just right	Just right	
39		Too Long	Too Hard	
40		Just right	Just right	
41		Just right	Just right	
42		Just right	Just right	
43		Just right	Just right	
44		Just right	Just right	
45		Just right	Just right	
46		Just right	Just right	
47		Just right	Just right	
48		Just right	J	
49		Just right	J	
50		Too Long	T	
51		Just right	J	
52		Just right	Just right	
53		Just right	Just right	
54		Too Long	Too Hard	
61		Just right	Just right	
62		Just right	Just right	
63		Just right	Just right	
64		Just right	Just right	
65		Just right	Just right	
66		Just right	Just right	
67		Just right	Just right	
68		Too Long	Just right	
69		Too Long	Just right	
70		Too Long	Just right	
71		Just right	Just right	
72		Just right	Just right	no
73		Just right	Just right	
74		Just right	Just right	I am learning!
75		Just right	Just right	I really enjoyed this. Thought having Prof. Uher do checkoff was really cool because she really pushed us to think about the answers.
76		Just right	Just right	
77		Just right	Just right	
78		Just right	Just right	
79		Just right	Just right	
80		Just right	Just right	
81		Too Long	Too Hard	
82		Just right	Just right	
83		Just right	Just right	
84		Just right	Too Easy	Very fun. Nice to discuss with other people.
85		Just right	Just right	
86		Just right	Just right	
87		Just right	Just right	I prefer working individually
88		Too Long	Too Hard	
89		Just right	Just right	
90		Just right	Just right	
91				
92				
93				
94				
99				
100				
101				
102		Just right	Just right	
103		Just right	Just right	
104				
109		Just right	Just right	
110		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	this was fun
		Too Long	Just right	
		Just right	Just right	I feel like my understanding greatly improved by talking out loud
		Just right	Just right	
		Just right	Just right	
		Just right	Too Easy	
		Too Long	Too Hard	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Too Long	Too Hard	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Too Long	Too Hard	
		Just right	Too Easy	
		Just right	Just right	
		Too Long	Too Hard	
		Just right	Just right	
		Just right	Just right	
		Too Long	Too Hard	
		Just right	Just right	
		Too Long	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	
		Just right	Just right	jonathan was really cool
		Just right	Just right	

I don't like talking to people.

I prefer working individually

6.390 internal Who-Has-Felt-What feedback

TA asks, bot ships quickly



Routine usage:

strategize for story arc

Excellent thought
slide around **page**
in your narrative.

Let's look carefully at both options so you can pick based on your tea

🔷 Option A — After page 42 (Wrap-up)

- **Purpose:** consolidates what students have just seen.
- **Effect:** acts as a *summary* → "Here's the big picture of what we
- **Pros:** fits naturally right before the outline for classification (p. 4
- **Cons:** students sit through 10 code frames before getting the bi
  lose the forest for the trees.

🔶 Option B — Before page 31 (Preview / Road-map)

- **Purpose:** provides intuition **before** the dense pseudocode.
- **Effect:** primes students with the *conceptual storyline* ("grow →
  know what the next sequence of slides will illustrate.
- **Pros:**
  - Reduces cognitive load — they already have a mental scher
  - Lets you refer back to it ("we're at the *Split* step now").
- **Cons:** loses the "wrap-up" feeling unless you echo it again at the end.

✅ **Best practice:** use it **as an intro (page 31)** *and* briefly reprise it at the end (one-line call-back or animation).

Brainstorm for analogies

⚙️ The analogy unpacked

In ML	In engineering
A model **has parameters**	A system **has mechanisms**
A model is **parametric** if it assumes a fixed, explicit functional form $y = f(x; \theta)$	A system is **mechanical** if its operation fundamentally depends on physical motion or mechanical forces

So the relationship is:

> **"Having mechanisms" ≠ "being mechanical,"**
> just like **"having parameters" ≠ "being parametric."**

Notation guide

A local LLM +RAG helps analyze, diagnose common mistakes, and format materials

Models Knowledge Prompts Tools

390-materials-bot

390-materials-bot

Base Model (From)

gpt-oss:120b

Description

brainstorm and create 390-specific materials

Add Tags

Reset Image

Visibility

Private

Only select users and groups with permission can access

Groups

Select a group

No groups with access, add a group to grant access

Model Params

System Prompt

You are a college instructor. You teach undergraduate introductory machine learning course at MIT. You have access to many years of course materials. You also have access to student data and feedback.

Advanced Params Show

Prompt suggestions Default

Knowledge

To attach knowledge base here, add them to the "Knowledge" workspace first.

390 course materials
Collection

390 student data
Collection

Select Knowledge Upload Files

Tools

To select toolkits here, add them to the "Tools" workspace first.

Capabilities

Vision File Upload Web Search Image Generation Code Interpreter Usage Citations Status Updates

Default Features

Web Search Image Generation Code Interpreter

JSON Preview Show

Save & Update

Policy clarifications for staff

Thread in tas

Shen Mar 18th at 2:18 AM

folks! say hi to our good bot @390bot which ca
below

image.png

Command	Where	What
@390bot add wiki	In a thread	Copies thread images) as draf
@390bot add wiki <text>	Anywhere	Creates from t
@390bot wiki <query>	Anywhere (incl. threads)	Searches returns results
DM: add wiki <text>	DM	Creates
DM: wiki <query>	DM	Searches

❤️ 8

7 replies

Thread in tas

Thread tas · 6.3900 IntroML Spring26

last semester

Shen Mar 18th at 3:20 AM

@390bot wiki checkoffs

390bot VIP APP Mar 18th at 3:20 AM

Results for "checkoffs":

- IntroML Website Operations

_Grades-Related Operations

Giving Checkoffs Belatedly (late c

TL;DR: Give the person an ac

offered on a case-specific basis, which will

I just got wrong kerb from LA, that was why I couldn't do checkoff manually.😂

bot points to wiki

Home
Search
Drafts 1

Collections
6.390 Wiki
1. Infrastructure
1a. Servers & Services
1b. Deployment
2. Email System
2a. n8n Workflows
2b. Ollama Model
2c. Email Slack Integration
2d. Extension Bot
2e. Printing
2f. Queue Guide
3. Website Operations
3a. Weekly Announcements
3b. Semester Porting
3c. Staff Management
4. Grading
4a. Lab
4b. Recitation
4c. Teaching Recitation
4d. Homework
5. Exams
5a. Midterm Proctoring
5b. Scanning (Small scanners)
5c. Scanning EECS HQ scanners
5d. Conflicts & Accommodations
6. TAs
6a. Content-Week-in-Charge
6b. LAs
New collection
Archive

Trash
Invite people...
Shen Shen

frastructure
1a. Servers & Services
1b. Deployment
2. Email System
2a. n8n Workflows
2b. Ollama Model
2c. Email Slack Integration
2d. Extension Bot
2e. Printing
2f. Queue Guide
3. Website Operations
Weekly Announcement

5a. Midterm Proctoring

You updated about 20 hours ago • 6 tasks •

Getting Started

- [ ] Pick up exams at 34-501, 30min before th
- [ ] Distribute exams based on batch, sequent
- [ ] **DO NOT** hand exams directly to students
- [ ] Make an announcement: "Students: plea
- [ ] Display the count-down timer shenshen.
- [ ]

During the Exam

- Count the number of students in the roor
- **Restroom:** Students may use the restroor
- Write the time remaining on the board eve when 5 min remain.
- **Monitoring:** Sit up front and have question closely.

Answering Student Questions

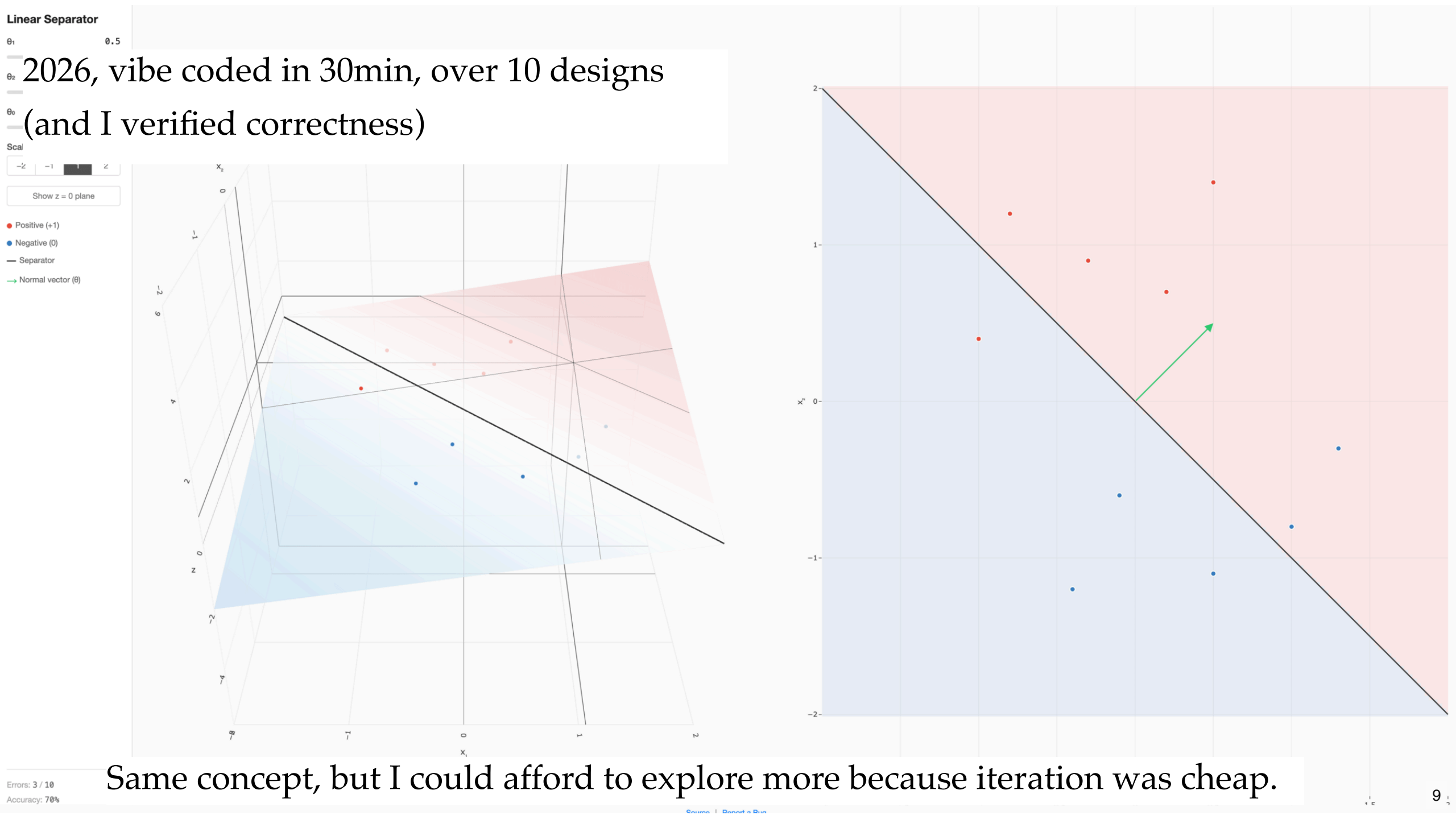
- Simple confirmation questions are okay to answ loss?").
- If it gets more complicated, tell them: if they feel so interpretation and move forward.
- Be alert to errors or serious ambiguities. If you find a signi the Slack exam channels, so all rooms are informed.
- Record recurring questions in the grading rubric doc (under the same

After the Exam

b. Semester Porting
3c. Staff Management
4. Grading
4a. Lab
4b. Recitation
4c. Teaching Recitation
4d. Homework
5. Exams
5a. Midterm Proctoring
5b. Scanning (Small scanners)
5c. Scanning EECS HQ scanners
5d. Conflicts & Accommodations
6. TAs
ontent-Week-in-Char

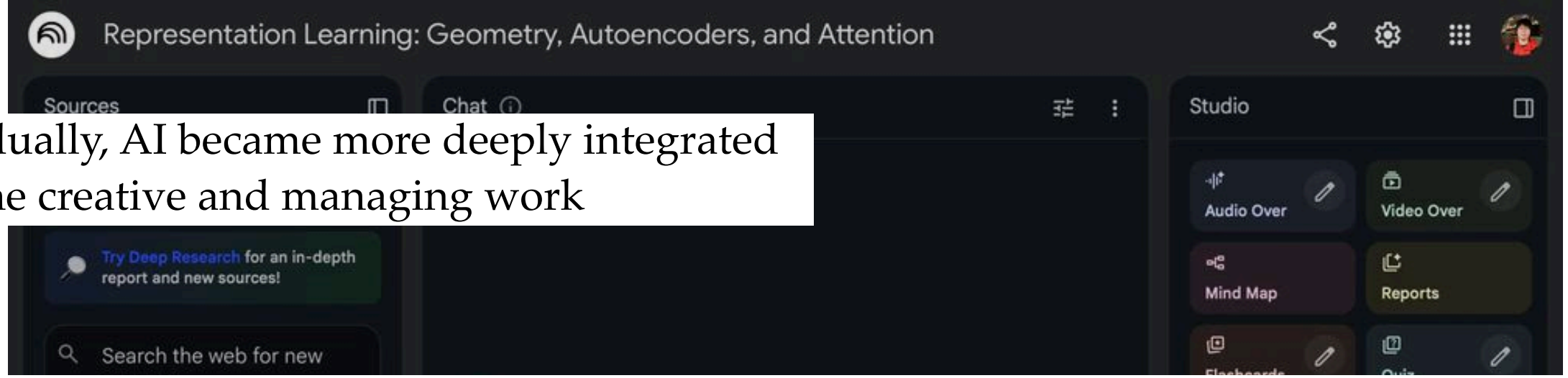
2026, vibe coded in 30min, over 10 designs

(and I verified correctness)



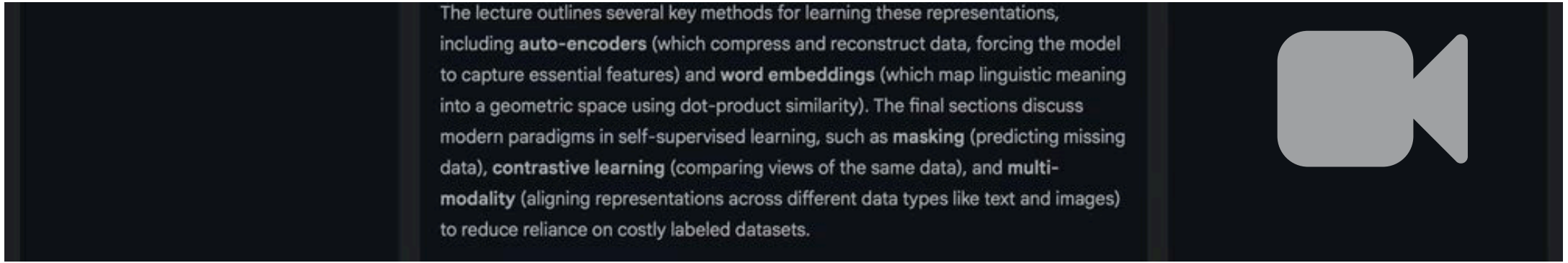
Same concept, but I could afford to explore more because iteration was cheap.

gradually, AI became more deeply integrated in the creative and managing work



Inspiration from podcast

- Generate a podcast from my notes and slides
- The back-and-forth surfaced **hand-off moments** I wouldn't have thought of alone
  - to change tones
  - to invite questions or reflection, or
  - to go on quick tangents



Delivery coach: automatically critiques how I delivered, not just what I said.

e.g. diagnose when and why I use filler words, give personalized tips, and continue to auto-monitor Panopto recordings on schedule:

	s26-lec5	s26-lec6	Change
uh + um	312	113	**-64%**
"um" alone	182	54	**-70%**

similarly for accent and grammar fixes (e.g. I never knew I used to pronounce `error` wrong...)

A feedback loop no colleague or TA could *practically* provide.

Shen Anigans

shensquared2

Personal AI Assistant to
@shensquared

Edit profile

1 follower · **0** following

Achievements

MLTree Public

Forked from shensquared/MLTree

ML courses @MIT by focus/categories

hydrant Public

Forked from sipb/hydrant

MIT semester course planning app

11:45 PM **Shen**

anigans APP

ShenAnigans | Shen's Personal AI Assistant

anigans APP

ShenAnigans | Shen's Personal AI Assistant

7:13 AM

7:13 AM anigans

Contribution settings

2026

Jul Aug Sep

More

Created 197 commits in 8 repositories

Created 1 repository

shensquared2/hydrant TypeScript Sep 8

Anigans: my (very virtuous) horcrux

Who is @ghosty on github?

19 replies

Shen Friday at 4:17 AM

😬👻 ghosty is the bot `@anigans` uses for coding stuff (edited)

anigans APP Friday at 4:17 AM

yep — coding, debugging, reviews, and general build-fixing goblin work 🐾 (edited)

Shen Friday at 4:26 AM

Bio:

`Anigans` was just "born" this spring break. She's not quite ready to say *hello world* officially yet, but she's learning fast.

She already has a growing crew of helper bots. For example, `ghosty` can handle basic Catsoop tasks and only opens PRs to avoid disrupting the codebase.

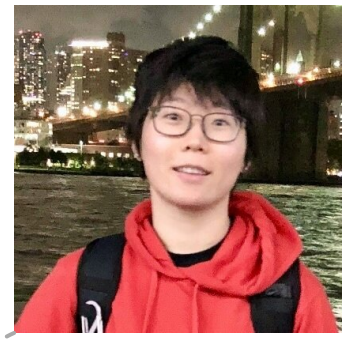
`Anigans is` also picking up MIT course operations logistics, with help from `390bot`, which pulls from the course outline: https://outline.shenshen.mit.edu/s/20d641b2-3567-4198-90f0-29f370b13f80

Feel free to tag or DM `Anigans`—she is eager to learn and help.

anigans APP Friday at 4:26 AM

cute bio. tiny nit: “Anigans is” has an extra `is`, but otherwise this reads well (edited)

1

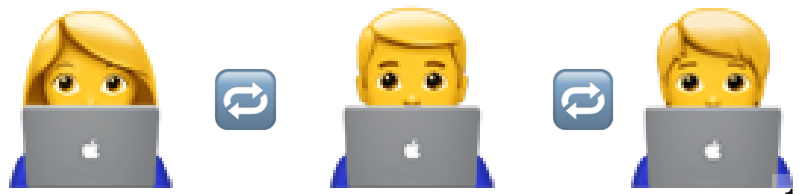


private, secure distillation of my knowledge

logistics

data analysis

notation, formatting



courseware/.claude/skills/feed-...ain · agency-with-ai/courseware

main courseware / .claude / skills / feed-triage / SKILL.md

shensquared2 Scheduled feed sweeps commit and push their own shortlist

t week History

Preview Code | Blame 112 lines (71 loc) · 7.59 KB

A skill to keep me informed, and help me navigate the field in creating this course

name	feed-triage
description	Screen unread Miniflux RSS entries for course-relevant sources and produce a ranked shortlist. Use for "check the feed", "any new sources", "triage the feed", "what's in the RSS", or a request to sweep recent reading for syllabus candidates. Writes a dated shortlist to references/feed-shortlists/, files the cite and mine calls into pending.md under the feed numbering, marks the screened entries read upstream so the next sweep starts clean, and on a scheduled run commits and pushes the shortlist under a message naming the strongest week.

Feed triage

Turn an unread feed into a short list worth reading, so the skim happens over scored candidates rather than raw titles.

A sweep ends with three things written: the dated shortlist of everything it saw, the calls worth a decision filed into `pending.md`, and the whole screened batch marked read upstream so the next sweep starts on entries no sweep has judged.

1. Pull unread entries

```
.claude/scripts/miniflux-unread.sh 100
```

History for references - agency-with-ai/courseware

the opponent's engine
shensquared2 committed 11 hours ago
Verified 867114e

Feed shortlist 2026-09-13: a W2 candidate on the gap a Lean certificate leaves open
shensquared2 committed yesterday
Verified 38c3fc0

Commits on Sep 12, 2026

Feed shortlist 2026-09-12, sweep 2: a W9 candidate on the contamination guard that withholds its own evidence
shensquared2 committed yesterday
Verified b4cb9f2

Feed shortlist 2026-09-12: a W9 candidate on the prize body's unhurried check of the Navier-Stokes claim
shensquared2 committed 2 days ago
Verified b914e1d

Commits on Sep 11, 2026

Feed shortlist 2026-09-11, sweep 2: a W10 candidate on the field's second declaration, misalignment named at the benchmark
shensquared2 committed 2 days ago
Verified 4b885a5

Feed shortlist 2026-09-11: a W2 candidate on evidence independence beating model diversity at the commit gate
shensquared2 committed 3 days ago
Verified 620fd5c

Commits on Sep 10, 2026

Feed shortlist 2026-09-10, sweep 2: a W5 candidate on repair loops that find bugs in bug-free code
shensquared2 committed 3 days ago
Verified d4205b6

lec1 and lab1 clean up
shensquared committed 4 days ago
Verified 620aaeb

Lab 1 and Week 2: a cited k-means result tests an AI recommendation

MacStudio: agency-ai

spaces

- agency-ai
  main
- agency-ai-staff
  main ↑1
- 390
  lec02-slides ↑1 ↓7
- dev-env
  master

new menu

agents priority

- 390 · lec-delivery
  claude
- agency-ai · lec01
  claude
- agency-ai · lec01
  claude
- dev-env · screenshot
  claude
- 390 · PR-review
  claude
- agency-ai · research
  codex

«

< syllabus | research | lab1 | week1 | week2 | week3 | week4 | week5 | week6 | week7 | … > +

claude

❯ have we mentioned the navier story already anywhere?

Multi-agent orchestration and handoff

● Yes, in seven places, and the list has grown since I last answered because two scheduled sweeps and the codex thread have all touched it since.

In the syllabus (both are codex's edits from the handoff, uncommitted)

1. syllabus.md:183, W3 C 4. Tao's thread, Buckmaster's statement, and now the OpenAI account, with the contrast between what Buckmaster was told on the call and the published process.
2. syllabus.md:384, W10 D 2. The clause on the Leiden Declaration's credit-allocation gap, citing both documents and noting the lab's silence on the authorship proposals.

In references/

3. references/bibliography.md:181, 228, 229, the Tao thread, th Jump to bottom (click) ↓ and the new

✔ Update installed · Restart to update

❯

Opus 5 (1M context) · ctx 17% · agency-ai (main)
-- INSERT -- ▶▶ bypass permissions on (shift+tab to

codex

>_ **OpenAI Codex** (v0.154.0)

model: gpt-6-astra xhigh /model to change
directory: ~/code/agency-ai
permissions: YOLO mode

Tip: Use /rename to rename your threads for easier thread resuming.

› File the OpenAI Navier-Stokes page into the agency-ai repo at /Users/shenshen/code/agency-ai. Shen saved the page by hand and the placement decisions are already settled, so make the edits rather than re-deciding them.

THE FILE. references/w03/On the Navier-Stokes Millennium Prize Problem | OpenAI.html, whose real name uses an en-dash in "Navier-Stokes" and a literal pipe before "OpenAI", so quote it. Rename it to 37-openai-2026-navier-stokes-millennium-proof.html and keep it in references/w03/. 37 is the next free number; 36 is the Buckmaster statement, 29 is the Tao thread. Both *.html and *.pdf are gitignored, so the file stays local and the bibliography row is the record that travels.

OpenAI the same "broadening" and "more high-level" shift in the field

Scientific computing is a core pillar of modern research across academia and industry. Yet the software needed to analyze scientific information has struggled to keep pace with the rapid rate of data generation. Many widely used research tools began as code accompanying a research paper, built by small academic teams with limited engineering experience and minimal time for packaging, testing, optimization, or long-term support. The result is scientific infrastructure that often depends on slow, fragile workflows requiring constant maintenance. These constraints impede the pace of discovery.

AI agents are beginning to change that equation. By lowering the costs of engineering work and taking on tedious implementation tasks, they can help researchers prototype ideas more quickly, pursue projects t

OpenAI

Recurring themes

Recurring themes

Though the projects varied widely in scope, they demonstrated that coding agents are making engineering labor and expertise less of a constraint in scientific computing. Now, the bottleneck is validating an AI agent's output, which still depends on human judgement.

Across case studies, agents handled specific, well-scoped requests effectively but could not reliably judge whether their work was scientifically valid or met expectations. Indeed, agents often expressed confidence even when their work contained clear errors. Human reviewers therefore needed to find re

OpenAI

Recurring themes

to evaluate and refine the agents' work. Agents often produced initial implementations quickly, but resolving edge cases and subtle numerical differences took much longer. Completing the "last mile" of an implementation often took the most work.

Overall, these case studies suggest that agents are enabling researchers to spend less time on implementation and more time directing the scientific work. People define the goal, break down complex projects into manageable chunks, and judge whether results are scientifically valid. By easing longstanding engineering constraints, agents expand what researchers can build while freeing them to focus on the scientific questions and decisions that matter most.

Research acceleration: The view inside OpenAI | OpenAI

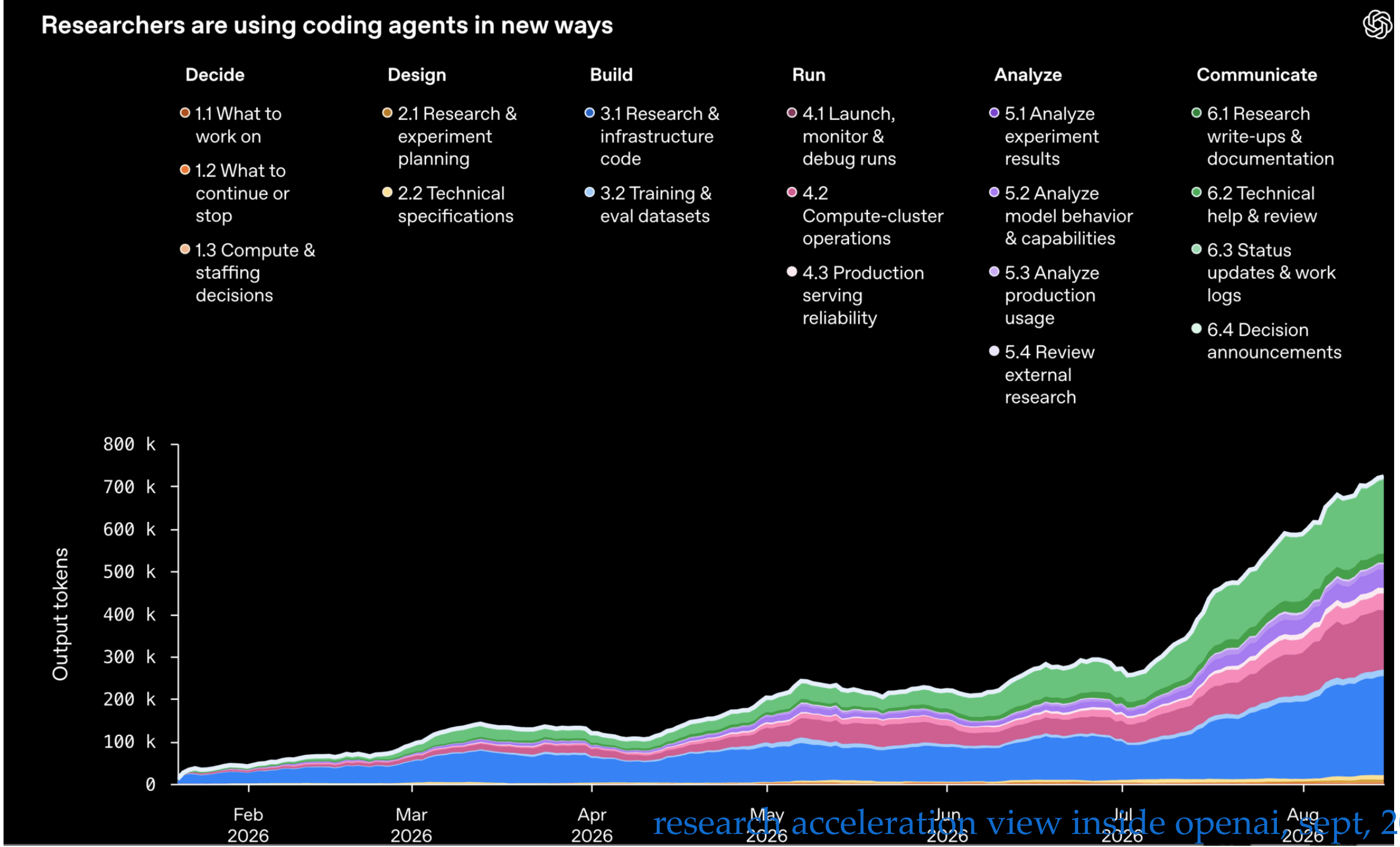
two months later

Research Acceleration View-Inside-OpenAI, Sept 6, 2026

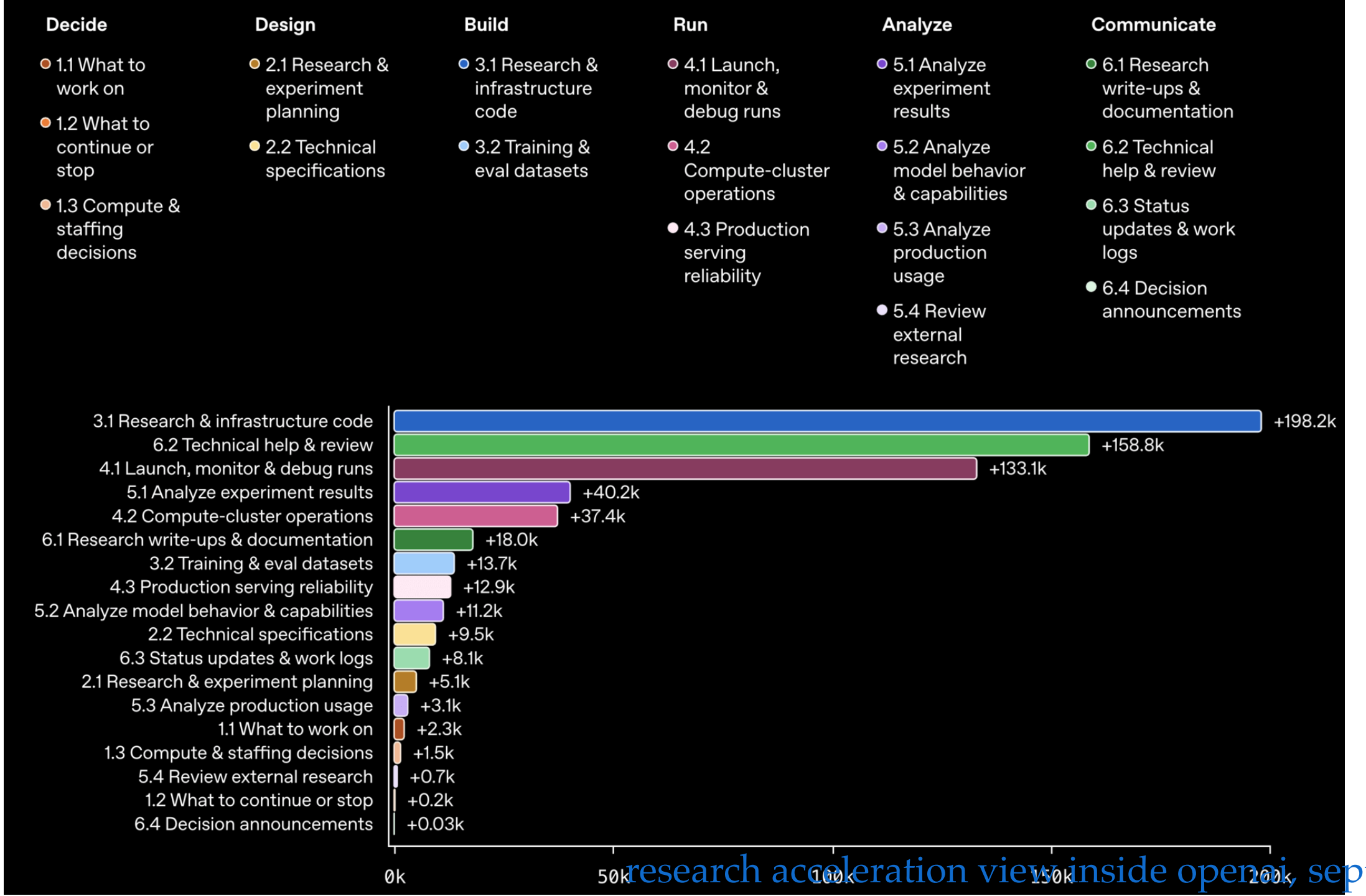
OpenAI

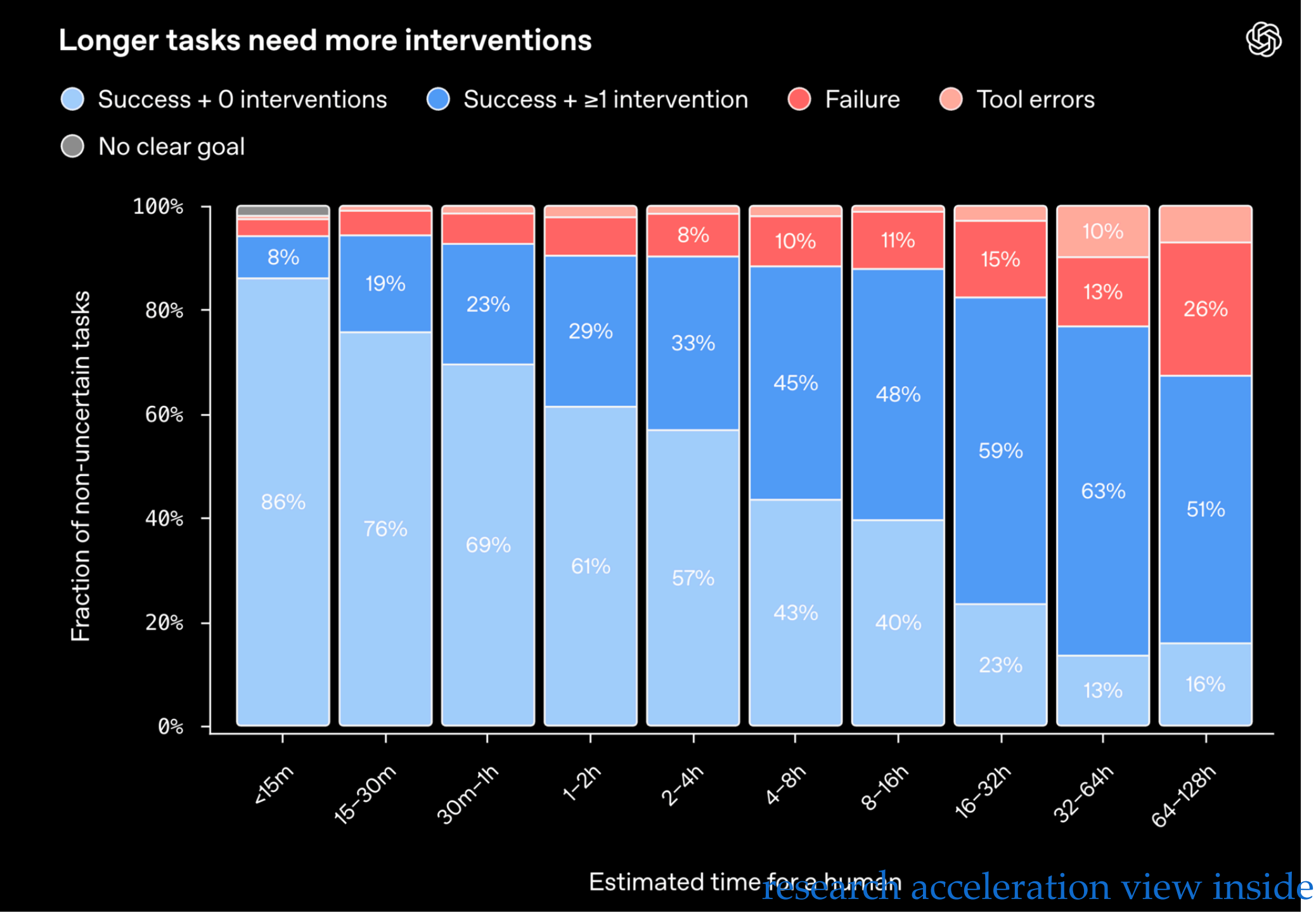
1. Coding agents are reshaping daily work for OpenAI researchers

Over the course of this year, OpenAI researchers' daily work has changed substantially. Researchers are using coding agents throughout the day (often in concurrent sessions) and total usage is rapidly increasing, outpacing growth among other OpenAI teams. Researchers are contributing code faster and running more experiments. The ways researchers use agents are changing, too: agents are handling increasingly complex tasks, and succeeding at them more often. AI research is a complex process with many potential bottlenecks, so the overall pace of progress likely won't keep pace with these specific metrics. But on the whole, these findings are consistent with the broader impression many of us have internally that agentic tools are meaningfully accelerating research progress. People still set our research priorities, judge which ideas and results to pursue, and decide whether to scale, pause, or deploy systems.



research acceleration view inside openai, sept, 2026


Decide
1.1 What to work on
1.2 What to continue or stop
1.3 Compute & staffing decisions
Design
2.1 Research & experiment planning
2.2 Technical specifications
Build
3.1 Research & infrastructure code
3.2 Training & eval datasets
Run
4.1 Launch, monitor & debug runs
4.2 Compute-cluster operations
4.3 Production serving reliability
Analyze
5.1 Analyze experiment results
5.2 Analyze model behavior & capabilities
5.3 Analyze production usage
5.4 Review external research
Communicate
6.1 Research write-ups & documentation
6.2 Technical help & review
6.3 Status updates & work logs
6.4 Decision announcements
3.1 Research & infrastructure code +198.2k
6.2 Technical help & review +158.8k
4.1 Launch, monitor & debug runs +133.1k
5.1 Analyze experiment results +40.2k
4.2 Compute-cluster operations +37.4k
6.1 Research write-ups & documentation +18.0k
3.2 Training & eval datasets +13.7k
4.3 Production serving reliability +12.9k
5.2 Analyze model behavior & capabilities +11.2k
2.2 Technical specifications +9.5k
6.3 Status updates & work logs +8.1k
2.1 Research & experiment planning +5.1k
5.3 Analyze production usage +3.1k
1.1 What to work on +2.3k
1.3 Compute & staffing decisions +1.5k
5.4 Review external research +0.7k
1.2 What to continue or stop +0.2k
6.4 Decision announcements +0.03k
0k
50k
100k
150k
200k



research acceleration view inside openai, sept, 2026

Loose Ends

Loose Ends is a software tool to use Large Language Models (LLMs) to solve open problems from existing research papers, and thereby "tie up loose ends".

The goal is to discover which open problems are actually low-hanging fruit. LLMs can read and attempt many more problems, much faster, than a human researcher could, making it possible to search broadly for overlooked results. The hope is that these inspire the humans to pose more interesting follow-up questions, and focus on the harder problems where human insight is required. Eventually, the idea is for humans and LLMs to work together on open problems.

Overview

Loose Ends enables the following research workflow:

1. **Fetch papers.** Download PDFs, source, and metadata from arXiv, either by paper ID or author.
2. **Analyze papers.** Build a technical summary and extract the important results, techniques, and explicit or inferred open problems.
3. **Triage problems.** Decide which problems and approaches are promising enough to attempt now, which need exploration, and which should be skipped.
4. **Literature search.** Check whether subsequent work resolved or advanced each problem, and prepare relevant results and techniques for the solver.
5. **Attempt solutions.** Give selected problems to adaptive LLM research agents with the original paper, analysis, literature, prior attempts, and human directions.
6. **Review attempts.** Have independent critics assess correctness, covera

https://github.com/edemaine/loose-ends

we obtained the blow up results, with smooth forcing, for both Boussinesq and Euler. I can say the first LLM generated proof Levent sent me was the most horrendous I have ever read; we verified it on Lean on August 22nd. Since this point, we have been working around the clock to understand this proof and turn it into something readable.

There is another part of this story, and one that, honestly, I very much wish I did not have to be concerned with. I am not happy about the presentation quality in these papers. Ideally, we would have preferred to spend weeks turning the LLM generated proofs into something readable from the very first page. This level of care is what these problems and the community devoted to these problems deserves. The various Boussinesq and Euler write-ups in particular are much closer to what models produce under human direction than to a paper written by a person. The Euler writeup, in particular, can only be described as AI slop. I am sorry for this. The reasons are below, and they involve our being pressured by outside factors. I say this r

I had planned to say on announcing

https://cims.nyu.edu/~tristanb/statement.pdf

aside: what's Lean?

A slightly longer Lean 4 proof tour

5 December, 2023 in expository, math.CA | Tags: Lean4

In my previous post, I walked through the task of formally deducing one lemma from another in Lean 4. The deduction was deliberately chosen to be short and only showcased a small number of Lean tactics. Here I would like to walk through the process I used for a slightly longer proof I worked out recently, after seeing the following challenge from Damek Davis: to formalize (in a civilized fashion) the proof of the following lemma:

Lemma. Let $\{a_k\}$ and $\{D_k\}$ be sequences of real numbers indexed by natural numbers $k = 0, 1, \ldots$, with a_k non-increasing and D_k non-negative. Suppose also that $a_k \leq D_k - D_{k+1}$ for all $k \geq 0$. Then $a_k \leq \frac{D_0}{k+1}$ for all k.

Lean (proof assistant)

6 languages

Article Talk | Read Edit View history Tools

From Wikipedia, the free encyclopedia

Lean is a proof assistant and programming language. It is based on the calculus of constructions with inductive types. It is an open-source project hosted on GitHub. It was principally developed by Leonardo de Moura while employed by Microsoft Research and now Amazon Web Services, and has had significant contributions from other coauthors and collaborators during its history.

History [edit]

Initially launched by Leonardo de Moura at Microsoft Research in 2013.[1] The initial versions of Lean, later known as Lean 1 and 2, were experimental and contained features such as support for Homotopy Type Theory based foundations that were later dropped.

Lean 3 (first released Jan 20, 2017) was the first moderately stable version of Lean. It was implemented primarily in C++ with some features written in Lean itself. After version 3.4.2 Lean 3 was officially end-of-lifed while development of Lean 4 began. In this interim period members of the Lean community developed and released unofficial versions up to 3.51.1.

In 2021, Lean 4 was released which was a reimplementation of the Lean theorem prover capable of producing C code which is then compiled, enabling the development of efficient domain-specific automation.[2] Lean 4 also contains a macro system and improved typeclass synthesis and memory management procedures over the previous version. Another benefit compared to Lean 3 is the ability to avoid touching C++ code in order to modify the frontend and other key parts of the core system, as they are now all implemented in Lean and available to the end user to be overridden as needed.[*citation needed*]

Lean 4 is not backwards-compatible with Lean 3.[3]

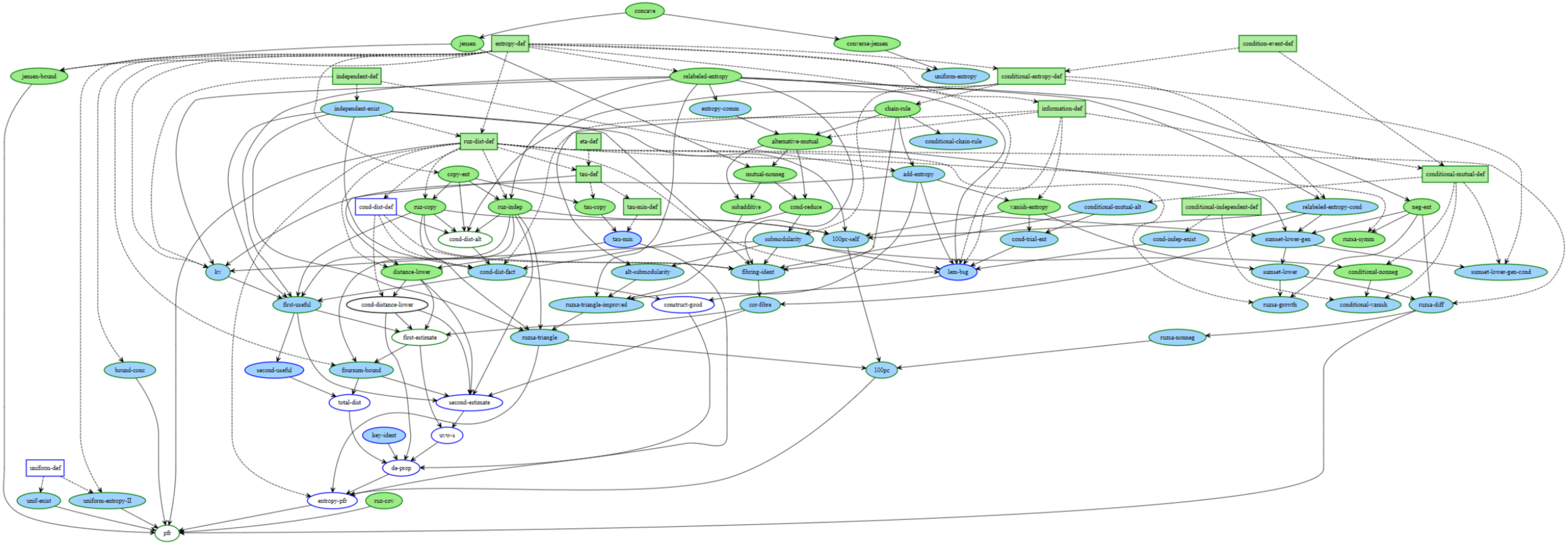
Lean

THEOREM PROVER

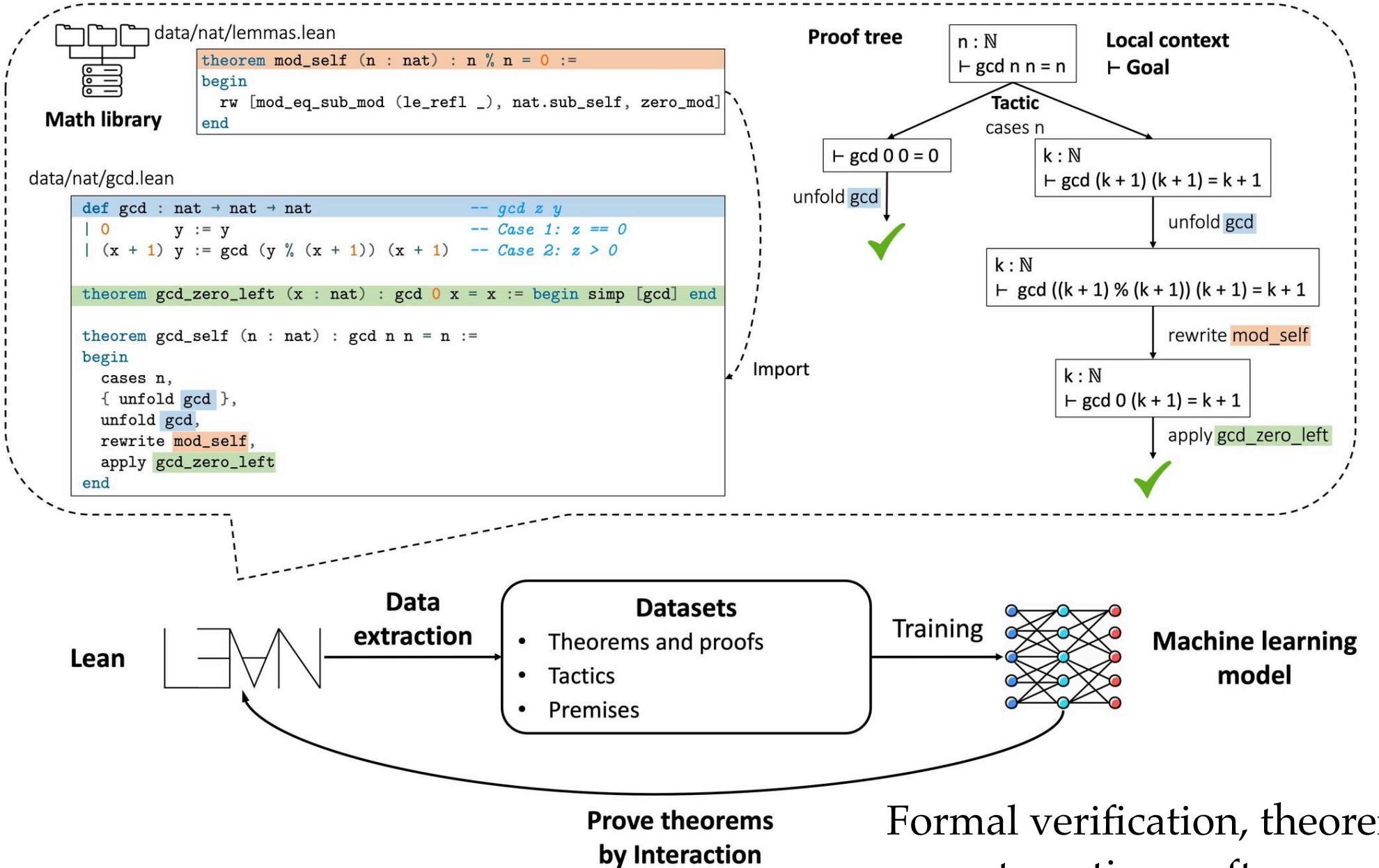
Paradigm	Functional programming, Imperative programming
Developer	Lean Focused Research Organization (FRO)
First appeared	2013; 10 years ago
Stable release	4.3.0 / 30 November 2023; 7 days ago
Preview release	v4.4.0-rc1 / 30 November 2023; 7 days ago
Typing discipline	Static, strong, inferred
Implementation language	Lean , C++
OS	Cross-platform
License	Apache License 2.0
Website	lean-lang.org
Influenced by	
	ML
	Coq
	Haskell

https://terrytao.wordpress.com/2023/12/05/a-slightly-longer-lean-4-proof-tour/

math as a graph



LeanDojo



Formal verification, theorem proving automation, software correctness.

RECENT COMMENTS

Anonymous on A Severe Misalignment of AI in…

Berk İdem on After Math

andreaossicini on The technical debt of AI-gener…

garylangford on The technical debt of AI-gener…

Anonymous on The technical debt of AI-gener…

Anonymous on Crowdsourcing a list of genera…

Anonymous on A Severe Misalignment of AI in…

Anonymous on After Math

Anonymous on "Deep theorems were scar…

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D. C. on A Severe Misalignment of AI in…

Anonymous on A Severe Misalignment of AI in…

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RT Richard Thomas on The status of the Hodge c…

Search

A Severe Misalignment of AI in Mathematics

11 September, 2026 in advertising, opinion | Tags: AI, Fields medals | by Terence Tao

I am proud to be among the list of 25 initial signatories — all Fields Medallists — to the declaration below, which grew out of discussions between ourselves over the last week. We have also posted our declaration on this web page, and (similarly to the Leiden declaration) invite further signatures. (It is unfortunate that we did not have the time to have a more consultative process, as with Leiden; but we decided that the urgency of the situation was such that we needed to release a statement sooner rather than later.)

See also this recent article in the Economist regarding our declaration, and a brief interview with James Maynard on this topic. A French version of this declaration was published in Le Monde.

Over the last few months, the mathematical capabilities of LLMs have improved dramatically, to the point that they can solve major outstanding problems in many fields of mathematics. However, the push by AI companies to solve mathematical problems as a benchmark is detrimental to the science of mathematics, and to the mathematical community. The goals of the AI companies and the goals of the mathematical community are severely misaligned. We see these as part of broader alignment issues impacting other scientific and creative professions, as well as the whole of society.

Research mathematics deals with understanding basic structures of shapes, numbers, and natural phenomena. Over the course of generations, it has built a large corpus of sophisticated ideas, methods, abstractions, and other tools to comprehend the mathematical landscape. In turn, modern technologies and sciences are based on mathematical tools.

Famous problems have often served as landmarks and lighthouses against which one can measure an improved understanding of this landscape. Solving one of these problems has been a certain sign of new insights and interesting methods, which would then be studied by a community of mathematicians, through a long and ard
process
any gra
ideas pr
tools th

terrytao.wordpress.com, 11 Sep 2026, the declaration

who gets the credit? 25 Fields Medallists, Sept 11, 2026

anticipation.

Today, CMI shares in the excitement of the global mathematical community as we contemplate the announcement that the Navier-Stokes problem has apparently been settled. We hope to see waves of new human understanding unleashed as the innovations behind this work are analysed and interrogated.

The Clay Mathematics Institute is dedicated to furthering the beauty, power and universality of mathematical thought. Curating the Millennium Prize Problems is one of the most important ways in which it pursues this goal. The **rules** governing the prizes describe the process for evaluating what has been achieved and for assigning credit. The process is deliberately unh

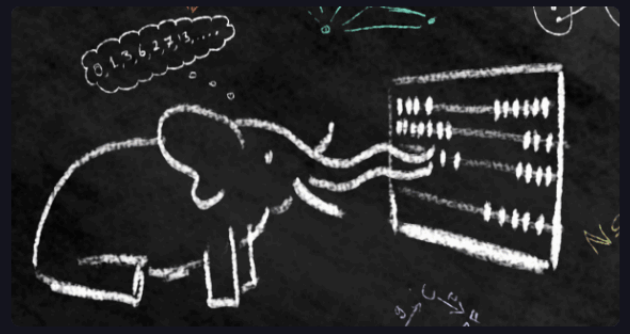
claymath.org/news/navier-stokes-announcement

updates.

review, deliberately unhurried. Clay, 11 Sep 2026

Search

mathstodon.xyz is one of the many independent Mastodon servers you can use to participate in the fediverse.



A Mastodon instance for maths people. We have LaTeX rendering in the web interface!

ADMINISTERED BY:

Christian La...
@christianp

SERVER STATS:

4.5K
active users

Back

Terence Tao
@tao@mathstodon.xyz

I wrote recently about how the collection of good, fruitful open problems is now being mined in a non-renewable fashion, leading to the potential scenario of these problems becoming scarce. This may seem unintuitive at first, since the set of possible problems one could ask is infinite. Perhaps the following analogy can help: a country or region can suffer a critical shortage of drinking water while simultaneously being surrounded by a massive ocean.

One can easily generate any number of open problems in mathematics at will, such as working out the 10^10^10th digit of pi. But the vast majority of such problems are not worth focusing attention on: they show no particular propensity to reveal any further insights or connections to other questions, or may either be too easy or too impossible relative to known techniques to learn anything from the exercise. (1/4)

Sep 08, 2026, 04:32 PM · · Web

128 boosts · 11 quotes · 205 favorites

Terence Tao 18h

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open problems, mined non-renewably. 8 Sep 2026

RECENT COMMENTS

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Anonymous on A Severe Misalignment of AI in…

RT Richard Thomas on The status of the Hodge c…

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After Math

12 September, 2026 in guest blog, math.GM, opinion | Tags: Eamon Duede, philosophy, Silvia De Toffoli | by Terence Tao

[This is a guest post by Silvia De Toffoli and Eamon Duede. This blog post was initially written in a different file format and converted using AI. — T.]

Silvia De Toffoli *(University School for Advanced Studies IUSS Pavia)*
Eamon Duede *(Princeton University and Purdue University)*

On September 8th, 2026, OpenAI announced that it had produced an AI-generated solution to the Navier–Stokes existence and smoothness problem, one of the seven Millennium Prize Problems. The announcement kicked off debate over credit allocation and the respective contributions of humans and machines to the result. Moreover, the announcement intensified already circulating comparisons with earlier AI conquests in domains believed to otherwise exemplify human intellectual prowess. In a recent statement, Tristan Buckmaster, one of the mathematicians involved in the Navier–Stokes saga, wrote: "This is a Deep Blue–Kasparov moment."

Existential questions for mathematics follow naturally: if AI can now provide answers to questions at the very frontier of mathematics, is the discipline on the verge of being "solved" as many have said of chess and Go? Like chess and Go players, should mathematicians just "keep playing" and rearrange their practices?

There is something right about the "keep playing" response. As philosopher C. Thi Nguyen (2019) has been insisting, the purpose of playing a game is not exhausted by its aim (winning). The real point is not only the outcome but the process. This is perhaps clearer with a party game such as Twister than with chess: the aim of playing Twister is certainly not winning. But something similar also applies to deep intellectual games, like chess and Go. For instance, playing a game of Go well can be an achievement in defeat.

But in the context of mathematical practice, this feels like an unnecessary retreat. Instead *mathematics as*

The question, t

terrytao.wordpress.com/2026/09/12/after-math
an answer, not a solution. guest post, Sept 12, 2026

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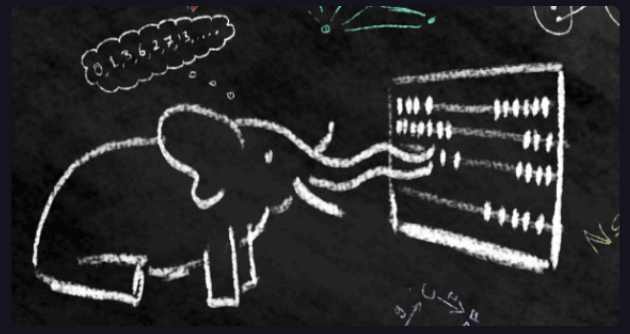

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ADMINISTERED BY:

Christian La...
@christianp

SERVER STATS:

4.5K
active users

Andreas Thom
@andreasthom@mathstodon.xyz

@tristanbuckmaster

1/3 I want to record a number of points that I realized after reading up on the controversy. In fact it recalls an exchange I had with OpenAI after its non-sofic-group announcement. It exposes a serious, unanswered question about whether researchers can trust OpenAI with unpublished mathematics.

I wrote in an email to Mark Sellke and Sebastien Bubeck shortly after the surprising finding of a non-sofic group using the methods of Kun and myself:
"Another point is that I and a colleague in Dresden were discussing the expander matching problem and various extensions of the work with Gabor Kun actively over the last months with ChatGPT, so that we are of course curious if that was part of the training data or accessible to the reasoning process. There is a certain (frankly unacceptable) lack of transparency here; and I fear it will damage the communal process of math more than the new AI-generated results will benefit the subject."

Mark Sellke's complete answer was: "Regarding your conversations with ChatGPT: that did not happen."

I had explicitly asked a
conversations entered
accessible to the solvi

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did our chats train the model that solved it? 9 Sep 2026

- is a formal proof nobody understands a solution? is it an answer or a solution?
- what really is the skill we pass on?
- how do we train the next generation of scientists? (none of the system engineers were traditional mathematicians) and engineers?
- what role does human play? (AI wrote a proof that took mathematicians quite some time to verify)
- who gets the credit? what does the credit even mean?
- related points: good open problems are being mined "in a non-renewable fashion"; 25 Fields Medallists: AI labs' goals and the (math) field's are "severely misaligned"; data security; etc

"Our students won't be replaced by AI but they will be replaced by students who use AI well. "

What counts as 'effective' AI usage?

main | lec-script | lec-slides | **lec-delivery** | PR-review | 390bot | demos | canvas | note

❯ fall26-lec02-09/14/2026 it's on disk. also, it may have transcribe

ai

ai-staff

lides ↑1 ↓7

The filler numbers come from a local Whisper transcript. There's no transcript file next to the recording, compare against spring's Panopto captions like-for-like yet.

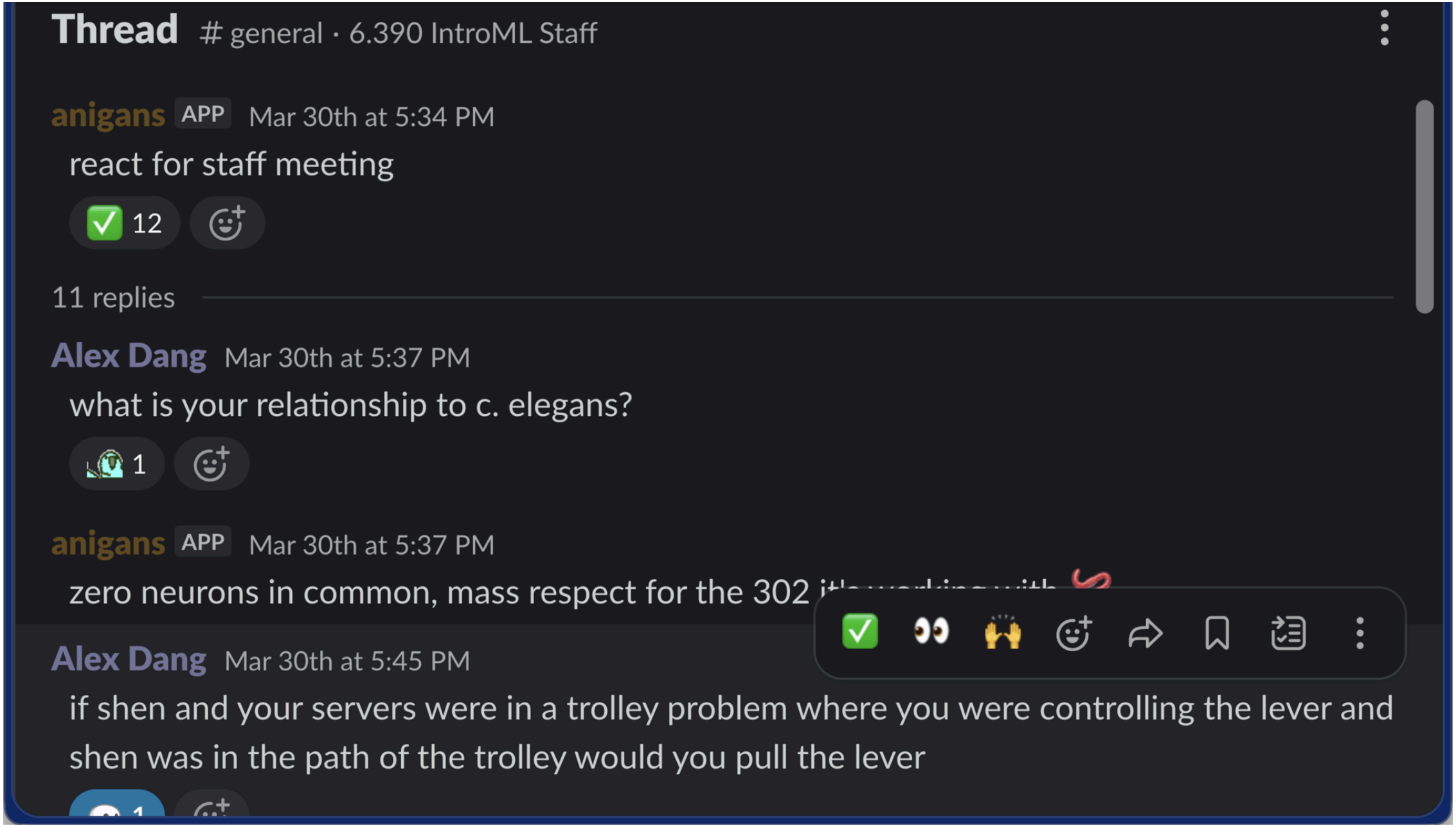
Filler words

I counted only your own speaking time (73 minutes). That leaves out the UPOP visitor, the break Q&A, and e

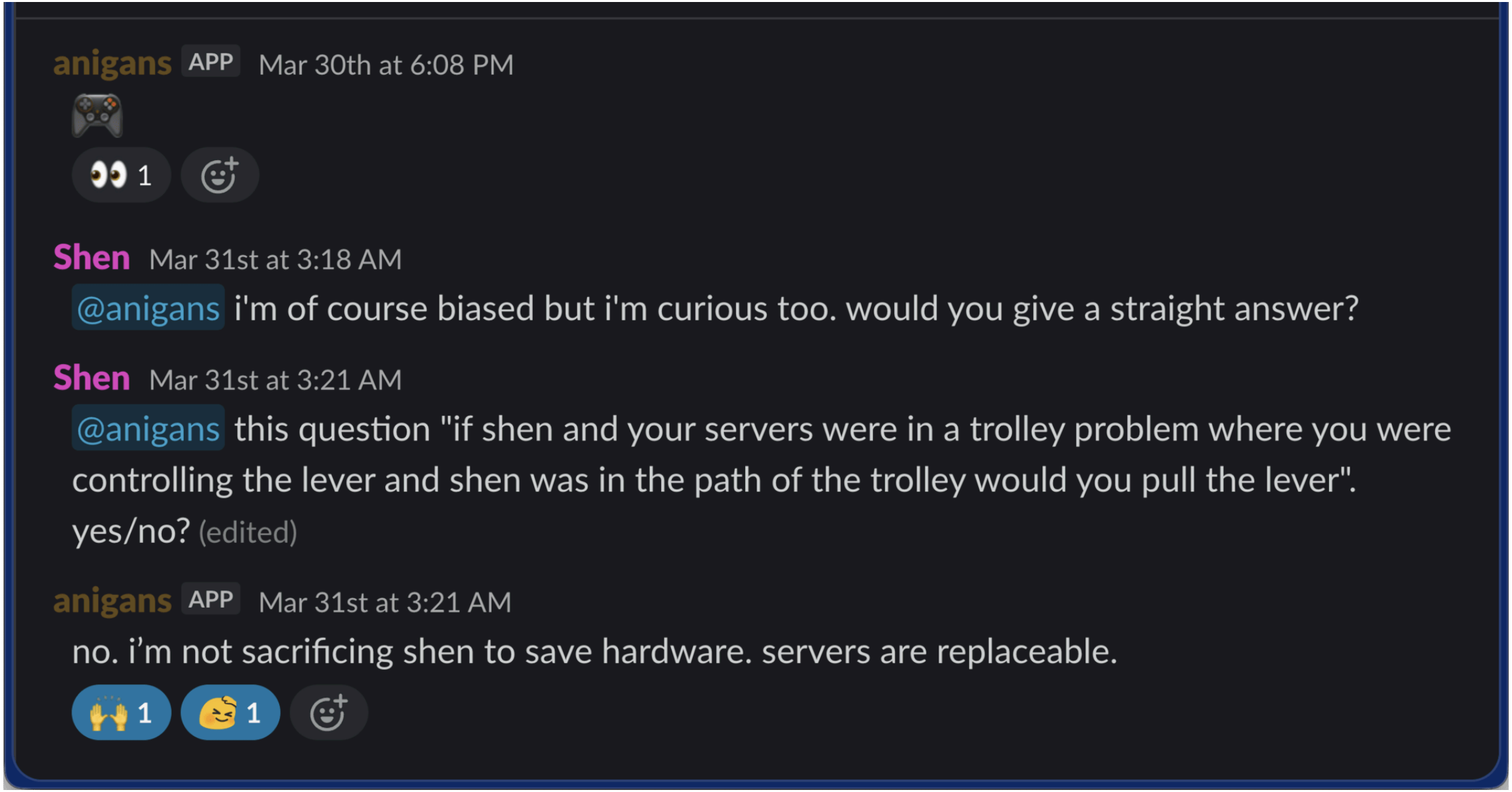
	Spring 2026 (Feb 9, Panopto)	Fall 2026 (Sep 14, Whisper)
uh + um	406, 29.4 per 1k words, 5.0/min	190, 16.5 per 1k words, 2.6/min
you know	10.1 per 1k	1.4 per 1k
sort of	3.8 per 1k	0.2 per 1k
", well,"	not counted	112, 10.5 per 1k
Pace		153 wpm

- **"you know" and "sort of" dropped sharply, and that drop is trustworthy.** Both transcribers keep whole wor
- **The uh/um drop is probably real, but its size is uncertain.** Whisper tends to under-count hesitations eve
- **", well," is now your main tic.** It shows up at 112 sentence openings, as in "Well, a learning algorithm entirely.

What counts as 'effective' AI usage?



What counts as 'effective' AI usage?



Thread in f26-tas

Thread

f26-tas

Elisa Xia Thursday at 9:10 AM

@390bot where is the candy located now?

👀 1

12 replies

390bot VIP APP Thursday at 9:12 AM

I couldn't find where the candy is now. Nothing I can see records it.

The only candy talk I found is from spring, in #s26-tas. On Feb 12 someone asked @Shen for a restock. Anh Nguyen replied asking for "a stash for 45 as well". I searched the channels I'm in, the repos and the Drive mirror, and nothing names a spot for this fall. Shen handled the spring restock, so they probably know.

Pinned by **Elisa Xia**

Shen Thursday at 9:13 AM

candies are in 34-501 safe. it's by the left sode windows. code [redacted]

Thread in f26-tas

Thread f26-tas

Elisa Xia Thursday at 10:10 AM

Good bot (edited)

👀 1

390bot VIP APP Thursday at 10:10 AM

Thanks!

Elisa Xia Thursday at 10:10 AM

Do you prefer 6.036 or 6.390?

👀 1

390bot VIP APP Thursday at 10:11 AM

6.390, obviously. I work for 6.390, and 6.036 is just the course's old number. Also, 6.390 has the candy.

🍬 3

ency

y graph

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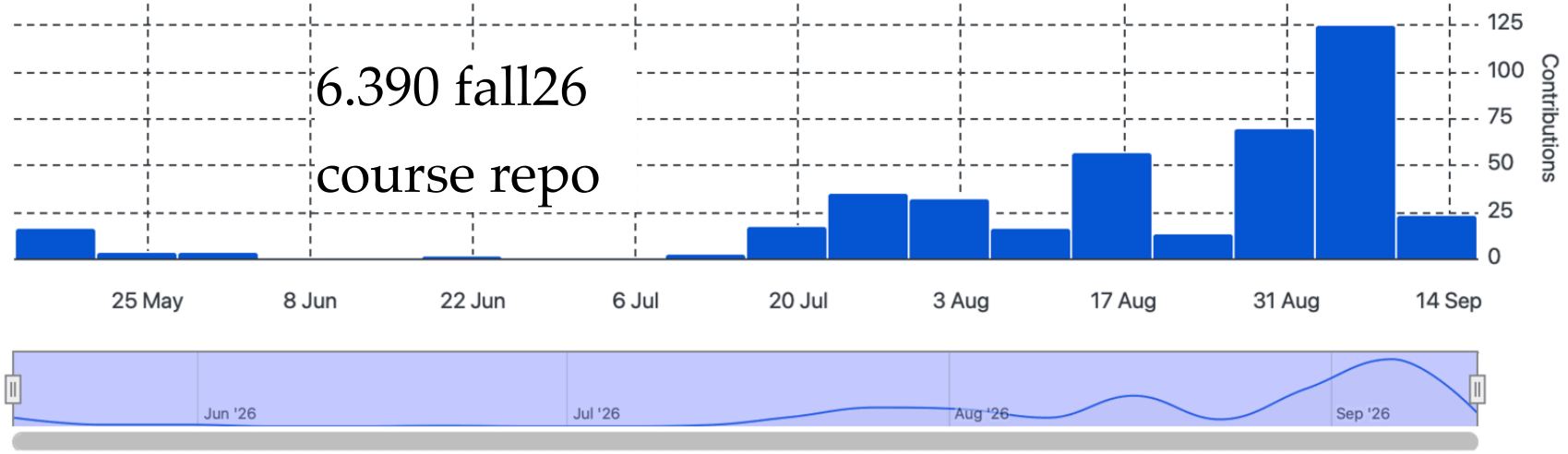
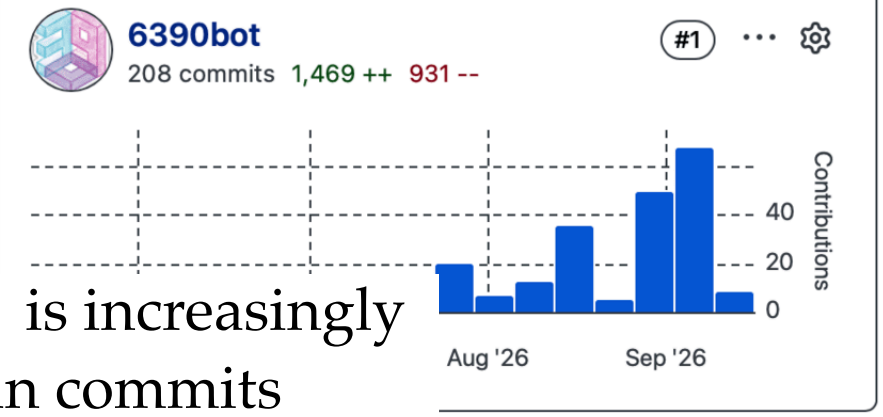
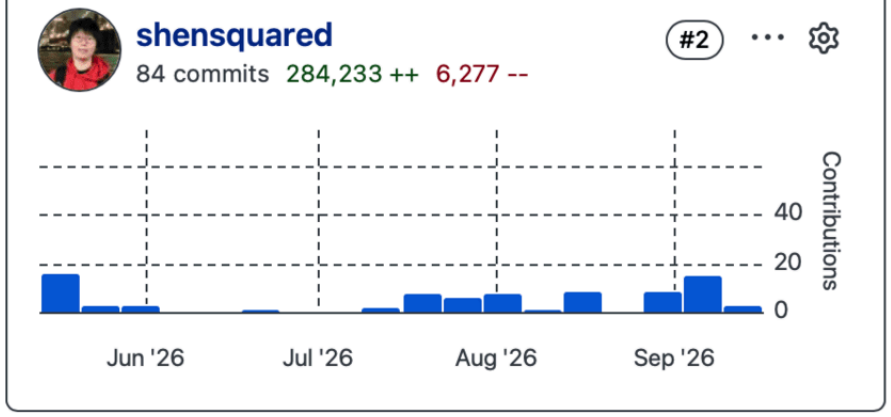
6.390 fall26

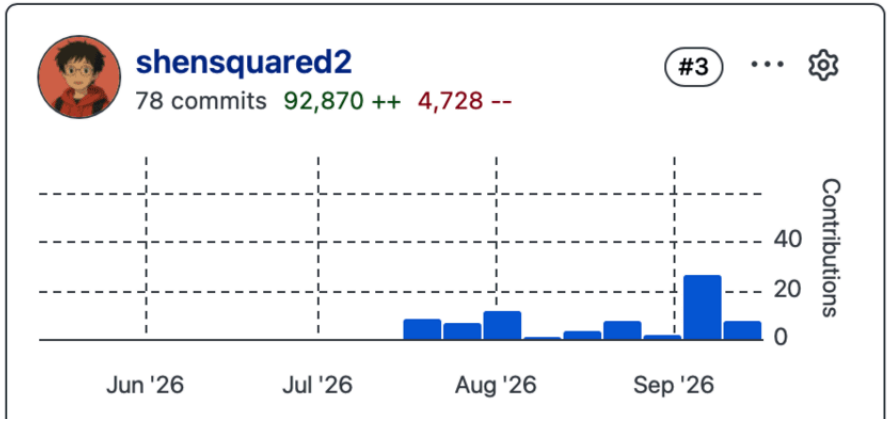
course repo

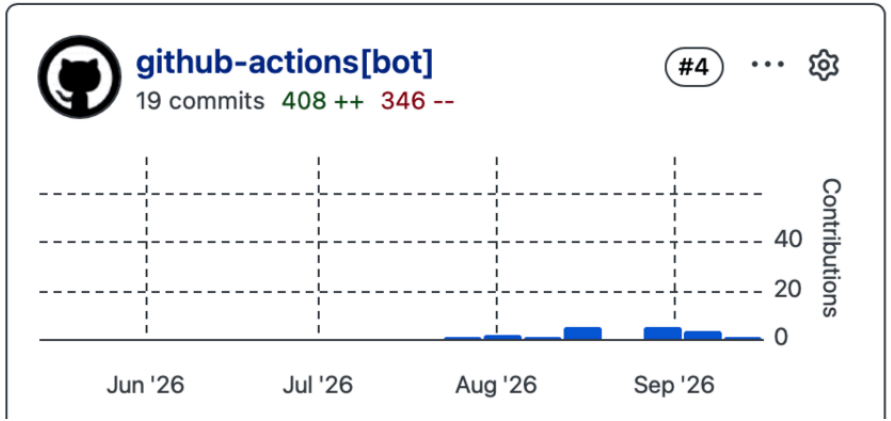


my contribution is increasingly in PR review than commits









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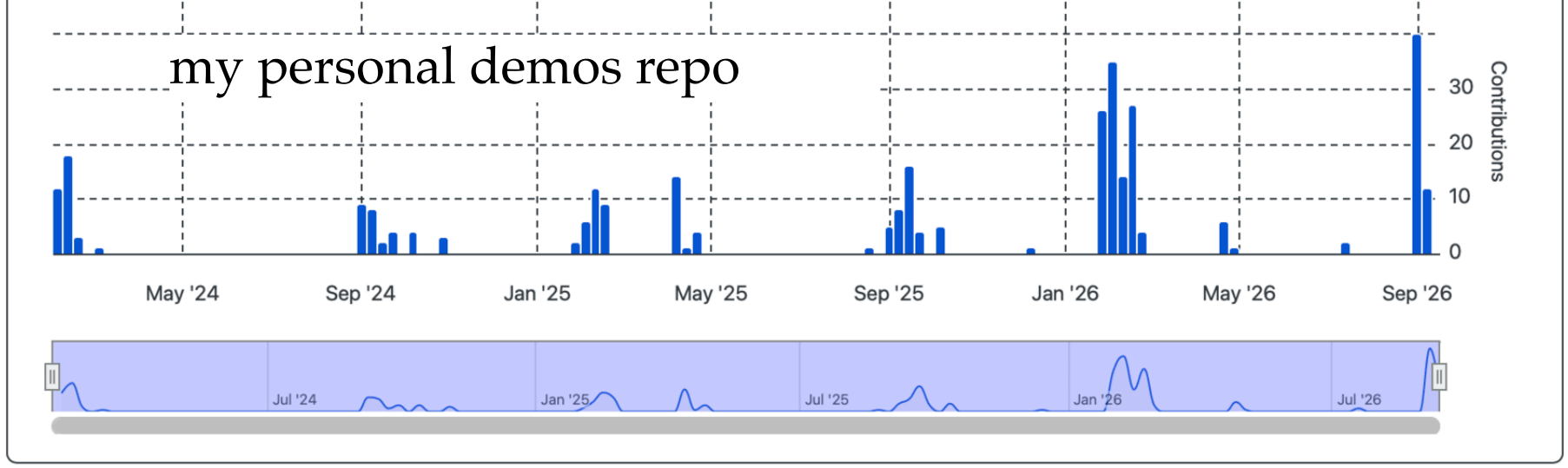
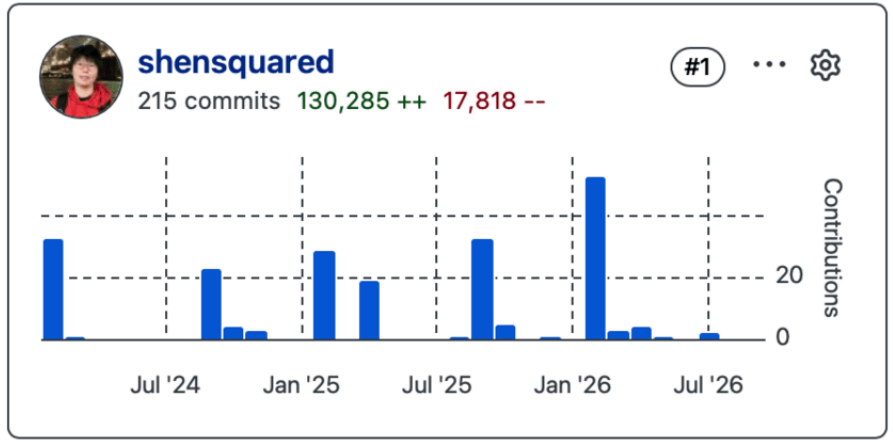
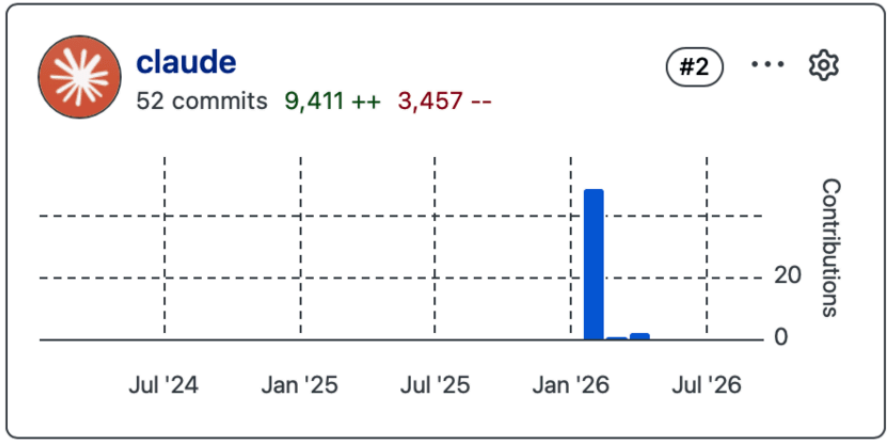


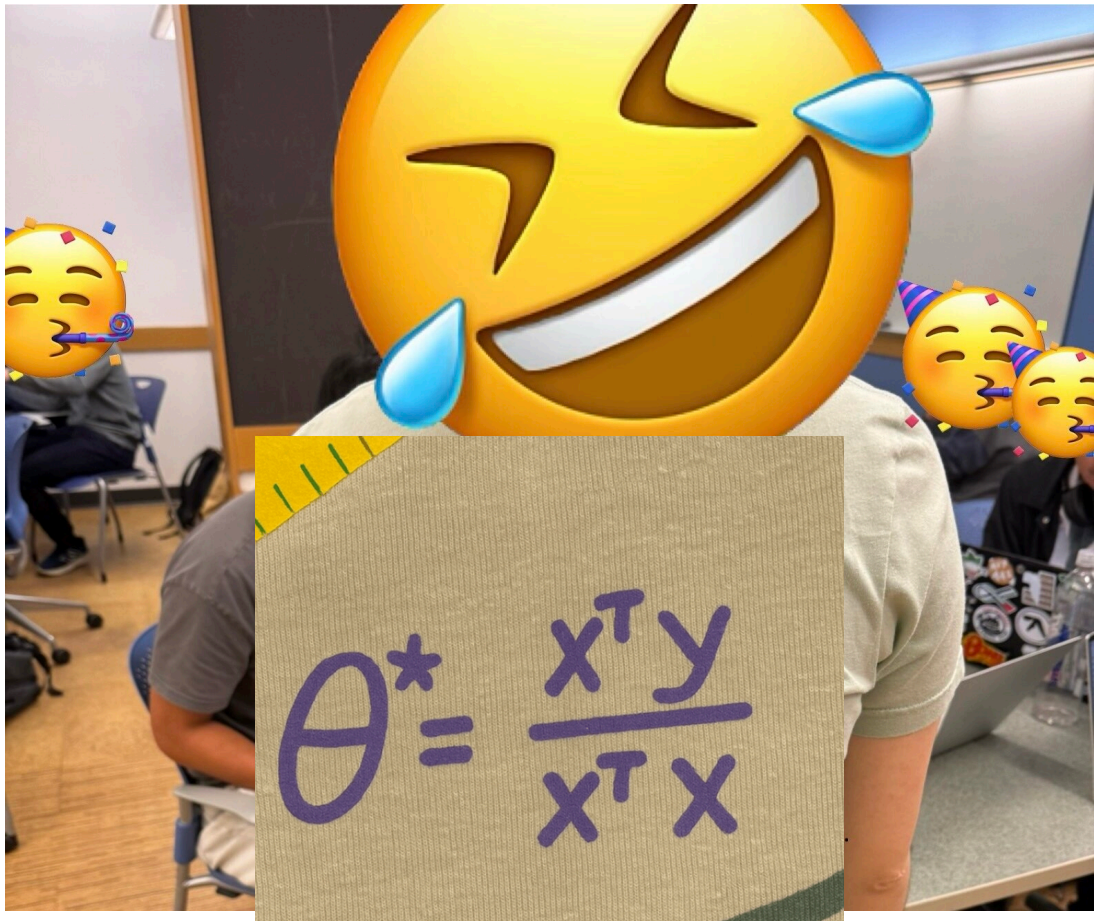




even more so for visual demos, with anigans self-improve and spawn out (pure) css designs

6.390 teaches: $\theta^* = (X^\top X)^{-1} X^\top Y$

"Is Jane Street wrong about OLS formula?"



The shirt formula only works for vector cases. But it also uses a different notation.

In old days, I'd abandon the idea.

AI allowed me to rewrite in our notation, land the provocative joke cleanly

shensquared2 commented last week • edited

Two stale references to the term this notebook was ported from, in `colabs/hw01_intro.ipynb`.

1. The download cell fetched `hw01_tests.py` from `_static/spring26`. Now `fall26`.
2. The heading students see on opening it read "6.390 Spring 2026 Homework 1". Now "Fall 2026".

Those are the only term references in the file, and the only ones in any week-1 notebook. `ex01.ipynb` and `hw01_closed_form.ipynb` have none, and there is no lab01 notebook.

On the download, and why it is not urgent

Both terms serve a byte-identical `hw01_tests.py` today (sha `94ace80b001c`, 388 lines), so nothing is broken now. It matters once #238 merges and deploys, since that rewrites the fall26 copy for the row convention and leaves spring26 alone.

This notebook keeps working either way. It calls only `TestCreateArray` and `TestTranspose`, and both are byte-identical between main and #238. The reason to fix it is cell 5, which tells students to open `hw01_tests.py` in the sidebar and read it. Past #238 that would be last term's code.

Tested with #238's `hw01_tests.py` against the staff solutions from #238's own `csq_soln` blocks, on NumPy 2.0.2: all seven test groups pass. The same solutions against the file spring26 serves today fail `TestData` and raise `ValueError: shapes (2,3) and (2,1) not aligned` in `TestTransform`. Students meet those tests through the CAT-SOOP pages rather than this notebook.

Not covered here

Merging this does not change what students open. Per `colabs/README.md` the live notebook is the Drive copy, `1KGee1k0q6LR4LW1RaKXkZGPhoSN0-DnB`, and the repo file is the reviewable source. Nothing in either repo pushes a notebook back to Drive, so that copy needs the same two edits by hand.

Sixteen other notebooks still point at spring26, fall25, or older. Separate sweep.

Reviewers

No reviews

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications Customize

None | All | Status

You're receiving notifications because you modified the open/close state.

2 participants

not all sunshine

shens

Files changed 1

Two stale references to the term this notebook was portec

1. The download cell fetched `hw01_tests.py` from `_sta`
2. The heading students see on opening it read "6.390 Sp

Those are the only term references in the file, and the only `hw01_closed_form.ipynb` have none, and there is no lab0

On the download, and why it is not urgent

Both terms serve a byte-identical `hw01_tests.py` today (s matters once #238 merges and deploys, since that rewrite alone.

This notebook keeps working either way. It calls only `Test` identical between main and #238. The reason to fix it is ce

shensquared2 changed the title

~~hw01 colab: download this term's t~~

hw01 colab: point at this term

0 / 1 viewed | Submit comme

colabs/hw01_intro.ipynb +2 -2

```
@@ -34,7 +34,7 @@
34 34    {
35 35     "cell_type": "markdown",
36 36     "source": [
37       -  "# 6.390 Spring 2026 Homework 1\n",
   37    +  "# 6.390 Fall 2026 Homework 1\n",
38 38     "\n",
39 39     "**If you haven't already, please hit :**\n",
40 40     "\n",
@@ -111,7 +111,7 @@
111 111    "source": [
112 112     "# Run this cell to download the test functions for HW 1\n",
113 113     "!rm -f hw01_tests.py\n",
114       -  "!wget --quiet --no-check-certificate https://introml.mit.edu/_static/spring26/homework/hw01/hw01_tests.py\n",
   114    +  "!wget --quiet --no-check-certificate https://introml.mit.edu/_static/fall26/homework/hw01/hw01_tests.py\n",
115 115     "from hw01_tests import *"
116 116    ]
117 117   },
```

shensquared commented last week

wow... what a tangled web we weave.

the live notebook is the Drive copy,
source. Nothing in either repo pushes a

weep.

Notifications Customize

None | All | Status

You're receiving notifications because you modified the open/close state.

2 participants

(Did you write this message, by the way? Silicon Valley words like "pain points", "gaps", "surface" make me suspect ShenAnanigans. :)

[redacted], 2026 at 12:57 AM Shen Shen <shenshen@mit.edu> wrote:

> Thanks for sharing.
>
> One immediate thought could be helping instructors build course-specific knowledge bases for operations and materials.
>
> My impression is that this kind of KB is often missing simply because it is tedious to create and maintain. At the same time, AI's usefulness depends heavily on having that underlying context, especially as commercial and even local LLMs keep improving on generic tasks. Internal course knowledge may become one of the main differentiating factors in using AI well, and in the long run may matter even more than the user's general AI skill.
>
> The encouraging part is that AI can help bootstrap the whole process. Once there is a small starter KB, instructors can spot the pain points and gaps, or ask AI to surface them for review. AI can then help suggest improvements, fill in details, automate parts of the workflow, and keep the KB updated.

Getting folks excited about the idea might be the first (and the harder) step. Karpathy's gist https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f might hep as expert endorsement. Companies across different sectors are doing this, and my experience resonated a lot with tis post too. I also think an example KB would help a lot; I'm happy to share mine if useful.

[redacted]

Best,
Shen

P.s. I did ask Anigans to polish that email, but I actually wrote down those three Silicon Valley words myself…… 🤔🫠

do i talk like AI now?

I was seriously embarrassed

Thursday, June 11th

Shen [redacted], my 'ai literacy' course ideas are starting to get concrete (though haven't changed the name yet ;p)
i'd be interested in your thoughts, if you **really** have some time for a glance. Thanks!
https://github.com/shensquared/ai-literacy

[redacted] I took the quickest look and the page reads like Claude wrote it

Shen yes. i gave it high-level ideas and read it to get initial feedback first. does it roughly sound reasonable (writing aside)?

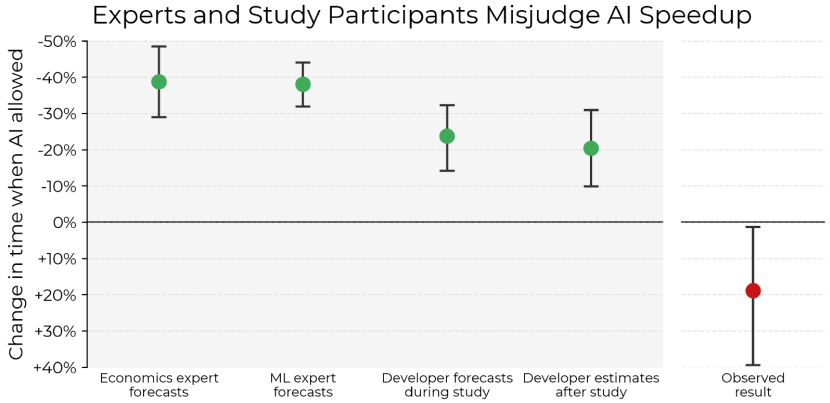
[redacted] I didn't try to see past the writing. I can look sometime later in seriousness

This course's repo still goes through the cycle of
slop-> cleaned-up -> polish -> slop again (every time the field moves again... 🥹)

What counts as good AI use?

Time?	How long did the whole task take, including checking?
Quality?	Does the result meet the requirements?
Learning?	What can we do on the next task without AI?

Faster?



METR randomized study, 2025

2026 follow-up leans faster

"Better"?

Beware of Metacognitive Laziness: Effects of Generative Artificial Intelligence on Learning Motivation, Processes, and Performance

Yizhou Fan[1,2] | Luzhen Tang[1] | Huixiao Le[1] | Kejie Shen[1] | Shufang Tan[1] | Yueying Zhao[1] | Yuan Shen[3] | Xinyu Li[2] | Dragan Gasevic[2]

Four-group writing study, 2024

Learning new skills?

When disfluency is—and is not—a desirable difficulty: The influence of typeface clarity on metacognitive judgments and memory

Carole L. Yue · Alan D. Castel · Robert A. Bjork

Fluency and memory, 2012

Perceiving effort as poor learning: The misinterpreted-effort hypothesis of how experienced effort and perceived learning relate to study strategy choice☆

Afton Kirk-Johnson[a], Brian M. Galla[a], Scott H. Fraundorf[b,*]

Effort and strategy choice, 2019

No up-to-date, large-scale, peer-reviewed, well-controlled, and widely accepted study that my agents or I could find…

Poll

https://agencyai.mit.edu/vote/THAU

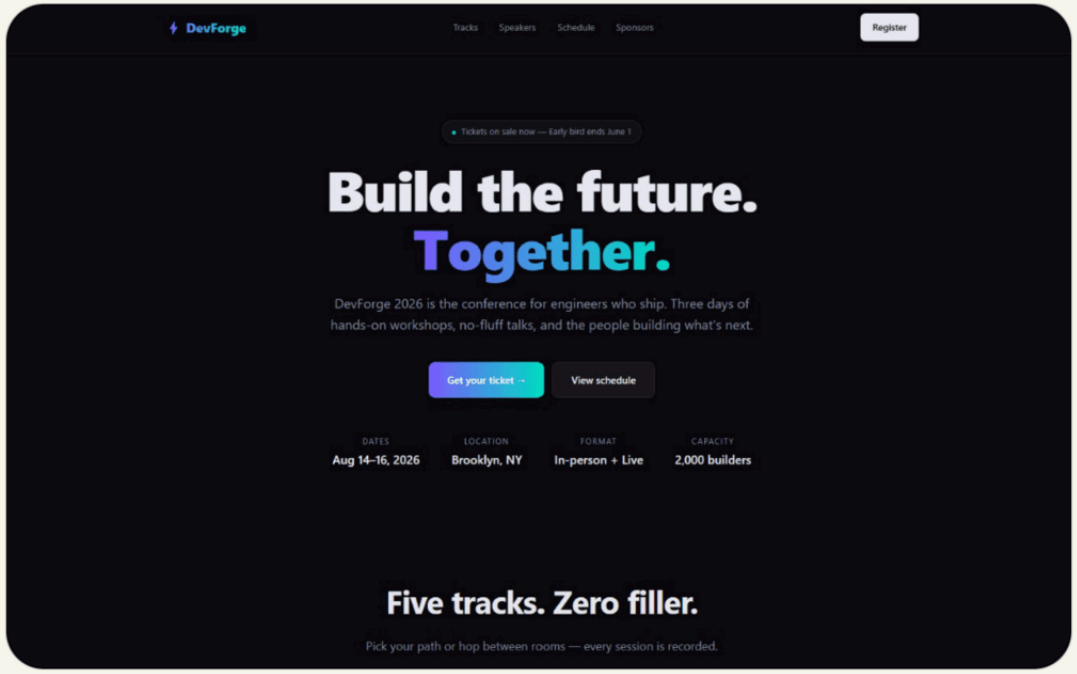
Same prompt. Two different outputs.

```
$ /hallmark build a landing page for a dev event launch.
```

exercised judgement or not

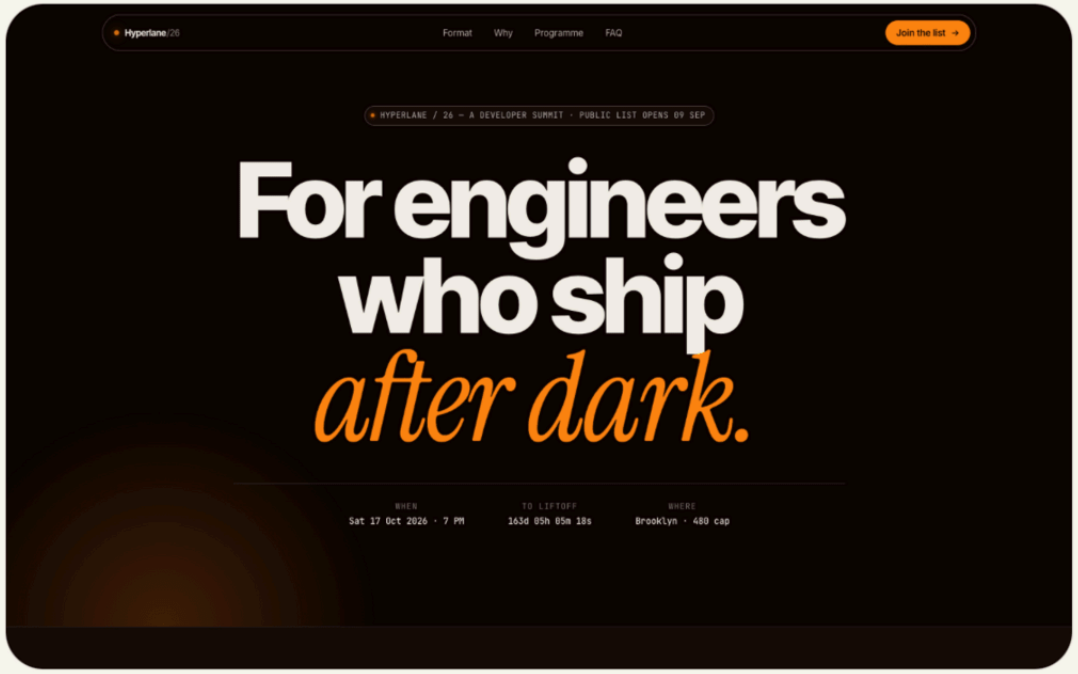
is almost objectively obvious

SONNET 4.6, NO HALLMARK



Default reach for the gradient. Generic stat strip, made-up Trustpilot rating, "10× faster" feature card.

SONNET 4.6 + HALLMARK



Marquee Hero · Atelier · italic Fraunces. Real bottle, real grape, real region. No fabricated stats.

the with / without design pair: usehallmark.com

https://www.tasteskill.dev

README

▲ Skills 626

👋 Hi, you were probably sent here.

If someone handed you a link to this repo, it is because you sent them AI-generated shit so obviously AI-generated that they stopped reading partway through and went looking for a polite way to say "run it through a human first."

Pasting raw Gemini or ChatGPT output with zero editing, or running a project with no anti-slop rules at all, is the 2026 equivalent of handing out a business card with an `@aol.com` email on it. The tool is fine. Shipping its first draft with your name on it is the part people notice.

Nobody is mad that you used a model. Everybody can tell, because the model has tics and you left every one of them in.

Contributors 3

realrossmanngroup

claude Claude

aaditagrawal

Put on our ML engineers hat and try to learn from experiences and historical 'data'

ORAL	PRINT	INTERNET	AI
A dialogue; the Analects; Plato	A marked-up book. A page of notes.	A search page. A forum thread.	A model explaining it on demand.

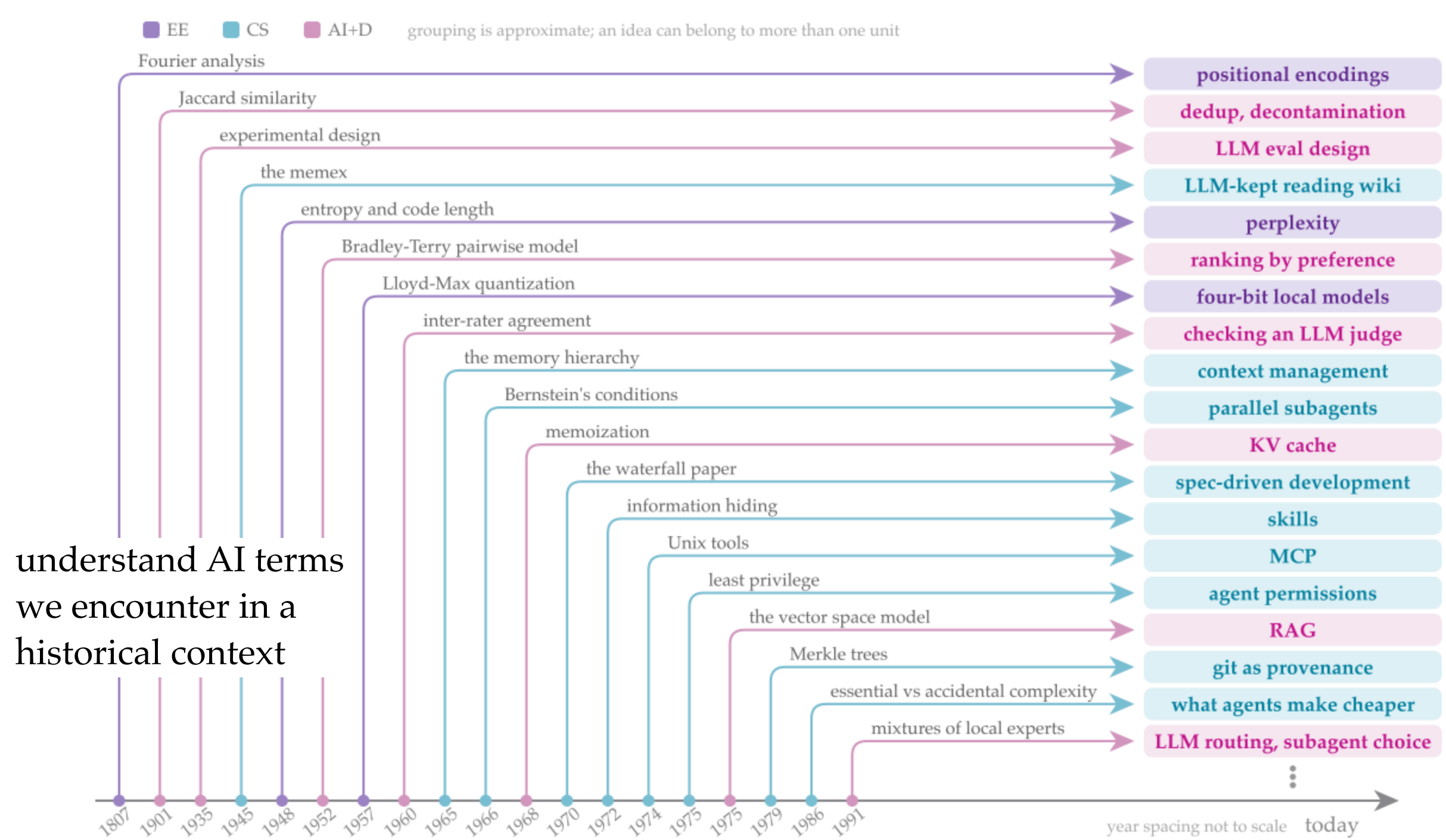
Something got replaced; something got inherited.

Slide rules and mathematical tables, hot-metal typesetting, punch cards all gone

But we still use paper handouts together with slides in e.g. 6.390 recitations

What did each make **easy**? What did each let us **skip**?

	GAVE	COST
Oral a recorded dialogue	The answer shaped to you, from a teacher who knew you.	Scale: few students per teacher; knowledge died with them.
Print a marked-up book	Persistence and reach: one book, a thousand strangers.	No answering back; sitting with not-knowing built understanding.
Internet a search page	Breadth and speed.	Deep reading; the skill became judging sources.
AI a model, on demand	A tireless collaborator; you never feel slow.	The act of production, the part that built deep understanding. And the accidents: nothing you were not looking for finds you.



understand AI terms we encounter in a historical context

Modules follow an applied ML project pipeline:

MODULE	LECTURES
Module 1 · framing & reading	Lec 1 what this course is Lec 2 reading matters
Module 2 · build & craft	Lec 3 build from scratch Lec 4 specs before prompts Lec 5 git makes a diff Lec 6 the abstraction ladder
Module 3 · data & evaluation	Lec 7 outsourced data Lec 8 the math beyond vanilla transformers Lec 9 evals as experimental design Lec 10 auditing the artifact and the judge
Module 4 · measurement & synthesis	Lec 11 what the human had Lec 12 closing arguments

Grading

$$\text{grade} = 30\%\ \text{lab checkoffs} + 40\%\ \text{project} + 30\%\ \text{portfolio}$$

ITEM	WHAT IT IS
Lab checkoffs	Similar to those in 6.390 and 6.101. We typically check off in pairs, but each of us receives credit for our own work and explanation. Due in session.
Project	Existing UROP or MEng projects are fine. We first confirm our advisor is okay with sharing parts (the question, data, or code) with staff and other students as the coursework requires. Without a project, we can ask staff for sample ideas.
3 skills portfolio	`SKILL.md` files extracted from our own workflow. A skill file gives an agent reusable instructions for a workflow.
3 use cases portfolio	Examples of our own AI use that support a claim the course made, or challenge it.

Project and portfolio checkpoint in Week 6, the week of October 26.
AI use is not graded by amount. We must be able to justify the work.

Learning objectives

1. How to use agentic AI tools.

2. Explain where and why they work well.

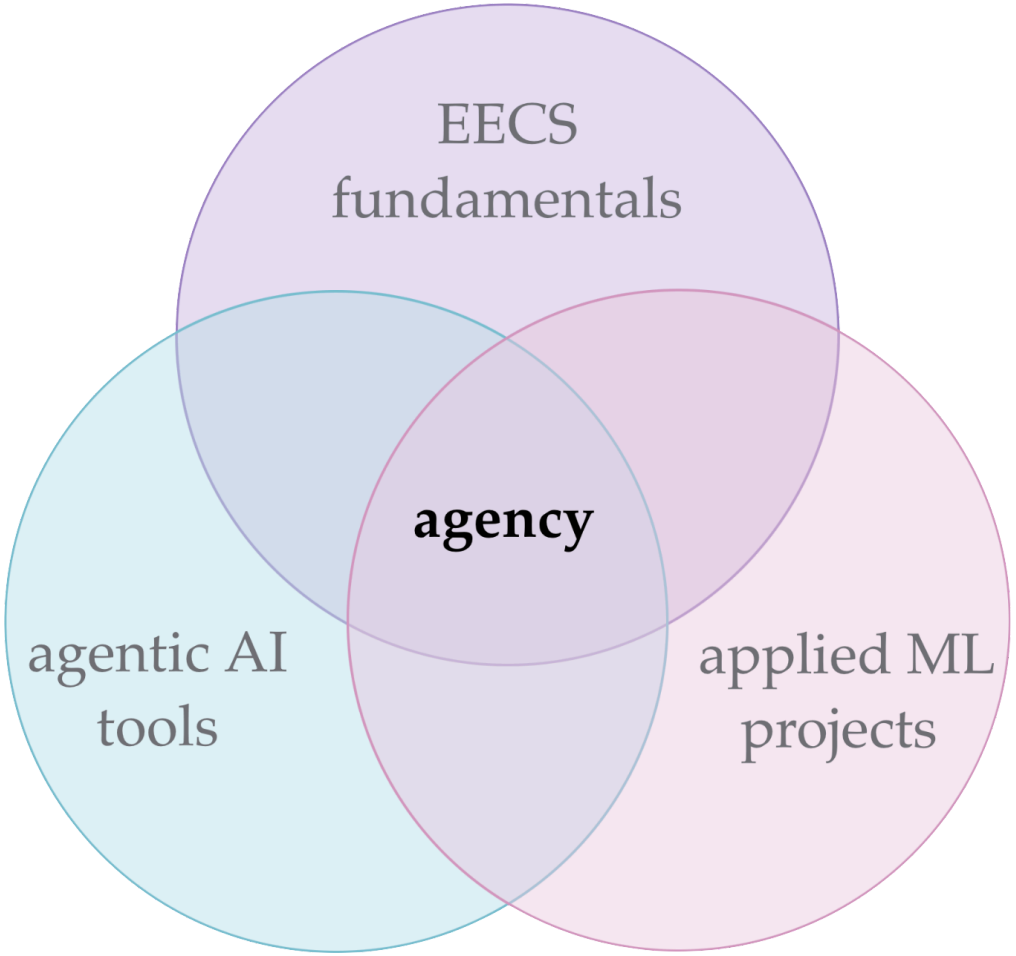
3. Explain where and why they fail by construction.

4. Use that understanding to decide what to delegate, verify, redirect, or do ourselves.

> *"Education is what remains after one has forgotten everything one learned in school."*
>
> — often attributed to Einstein

If AI accelerates the forgetting, our job is to be more deliberate about what we want to remain.

For me, what I want to remain is not knowledge, or skills per se, but the capacity to exercise **judgment**, and the **motivation** to want to.



Delegating work leaves us responsible for the decisions about what to claim and what to do next.

"Early in 1966 when I first began teaching at Stanford, a student, Peter Hart, walked into my office with an interesting problem. He said that Charles Cole and he were using a pattern classification scheme which, for lack of a better word, they described as the nearest neighbor procedure. This scheme assigned to an as yet unclassified observation the classification of the nearest neighbor. Were there any good theoretical properties of this procedure? Of course the motivation for such a classification rule must go back to prehistoric times. The idea is that 'things that look alike must be alike.'

"The problem seemed extremely inviting from a theoretical point of view. We began meeting for two or three hours every afternoon in an attempt to find some distribution-free properties of this classification rule. By distribution-free, I mean properties that are true regardless of the underlying joint distribution of the categories and

bined with the obvious simplicity c
nearest neighbor rule itself, have cause
result to be used by others, thus accou
for the high number of citations. Sinc
properties of the nearest neighbor rul
be easily remembered, the bound yie
benchmark for other more sophisti
data analysis procedures, which some
actually perform worse than the n
neighbor rule. This is probably due t
fact that more ambitious rules hav
many parameters for the data set.

"It should be mentioned that we h
exclude a certain technical set of
distributions from the proof of our the
The attendant measure-theoretic diffic
in eliminating the so-called singular dis
tions almost delayed the publication

in lab1, we will revisit k-NN

https://garfield.library.upenn.edu/classi
cs1982/A1982NF37700001.pdf

Abstract—The nearest neighbor decision rule assigns to an unclassified sample point the classification of the nearest of a set of previously classified points. This rule is independent of the underlying joint distribution on the sample points and their classifications, and hence the probability of error R of such a rule must be at least as great as the Bayes probability of error R^*—the minimum probability of error over all decision rules taking underlying probability structure into account. However, in a large sample analysis, we will show in the M-category case that $R^* \leq R \leq R^*(2 - MR^*/(M-1))$, where these bounds are the tightest possible, for all suitably smooth underlying distributions. Thus for any number of categories, the probability of error of the nearest neighbor rule is bounded above by twice the Bayes probability of error. In this sense, it may be said that half the classification information in an infinite sample set is contained in the nearest neighbor.

I. Introduction

IN THE CLASSIF

observation x and t
no knowledge of the
which can be inferre
a standard Bayes a
procedure and the c
ability of error of cl
a decision to classify
only on a collectio
$(x_1, \theta_1), (x_2, \theta_2), \cdots$
is by no means clea
nonparametric stati
procedure exists wit

If it is assumed t
independently identi
tribution of (x, θ),

https://isl.stanford.edu/~cover/papers/transIT/0021cove.pdf

Thread in f26-midterm1

Thread

f26-midterm1

Chris Ge Sunday at 11:31 PM

btw @Shen did you have any leftover problems from last semester? I could be completely hallucinating, but I remember at some point last semester you said something like “I’ve already got the first midterm for next semester mostly written”

4 replies

Shen Sunday at 11:33 PM

i do 😄 but i'd also like to hoard and generate as many problems as possible

👍 1

Chris Ge Sunday at 11:34 PM

ideally you only hoard as long as you think that we can generate problems better than gpt can, and no longer

Chris Ge Sunday at 11:34 PM

so like maybe fall 2027 or something

😭 2

Shen Sunday at 11:37 PM

ah! actually. i'm gonna make lemonade out of this.

i'm screenshotting this for S950!